

used in each individual factory will depend on local circumstances.

Treatment processes may also be described in the following manner:

1. Primary treatment; physical and chemical methods, e.g. sedimentation, coagulation etc.
2. Secondary treatment; biological methods (e.g. activated sludge) conducted after primary treatment.
3. Tertiary treatment; physical, chemical or biological methods (e.g. microstrainers, sand filters and grass plot irrigation) used to improve the quality of liquor from previous stages (Forster, 1985).
4. Sludge conditioning and disposal; physical, chemical and biological methods. Anaerobic digestion is often used to condition (make it more amenable to dewatering) the sludge produced in previous stages. Following dewatering (e.g. by centrifugation using a decanter centrifuge) the sludge can then be disposed of by incineration, landfilling, etc.

Physical treatment

The removal of suspended solids by physical methods before subsequent biological treatment will considerably reduce the BOD of the resulting effluent. In nearly all fermentation processes the cells are separated from the liquid fraction in recovery processes (Chapter 10). Obviously, biomass processes need not be considered. Yeast cells from other processes may be a marketable product, but microbial cells may not always be marketable, particularly when contaminated with filter aid. In these instances, when the cells and filter aid are a waste, the recovered material may be dealt with in two basic ways:

1. The waste is disposed of without any further treatment.
2. The waste bulk is reduced by mechanical dewatering with a filter press, centrifuge, rotary vacuum filter or belt press. The compressed waste is then incinerated (Grieve, 1978) or disposed of in a landfill site.

Solid wastes are produced in some processes before inoculation. In breweries, where malted grain is still used, coarse screens or 'whirlpool' centrifuges may be used to remove spent grain from the wort after it is mashed. About 5 kg (wet weight) of grain are produced

per barrel (180 dm^3) of beer. If hops are used, rather than hop extracts, they will also be recovered on screens in a 'hop back'. This residue may amount to 250 g per barrel. Both the spent grain and hop waste may then be mechanically dewatered before being sold or dumped.

The stillage (after distillation) in whisky distilleries may be passed through screens (1 mm openings). These screenings are then removed, mechanically dewatered, and dried in rotary driers to yield a potentially marketable residue known as Distillers' grains. According to a survey in Scotland, about half the whisky distilleries were evaporating the spent waste to a syrup containing 45% solids, mixing with spent grain, drying and selling the final product, 'Distillers' Dark Grains', as a low-grade cattle food (Mackel, 1976).

Physical processes installed for primary effluent treatment may include the following stages:

1. Screens, to remove larger suspended and floating matter.
2. Comminutors, to reduce particle size.
3. Constant velocity channels ($\sim 0.3 \text{ m s}^{-1}$) for grit removal to prevent damage to plant in later processes.
4. Sedimentation tanks for the removal of finer suspended matter. These are generally circular or rectangular continuous flow tanks operating at retention times of 6–15 hours (and designed to have a minimum retention time of 2 hours), with facility for the continuous removal of settled sludge. Sedimentation tanks can remove 70% of the incoming suspended solids and, depending on the nature of the waste, up to 40% of its BOD load (Forster, 1985). They can be operated with or without prior chemical coagulation/flocculation. Similar settlement processes are also conducted after secondary (biological) treatment.

Physical processes used in tertiary treatment to produce an effluent of better quality than the 30:20 standard include microstrainers, slow sand filters, upflow sand filters and rapid gravity sand filters. Throughputs vary between around $3 \text{ m}^3 \text{ m}^{-2} \text{ day}^{-1}$ for slow sand filters and $700 \text{ m}^3 \text{ m}^{-2} \text{ day}^{-1}$ for microstrainers. Suspended solids removal is generally 50–70% and BOD removal around 30–50%, depending on the technique used. A detailed description of tertiary treatment is given by Truesdale (1979) and Viessman and Hammer (1993).

Chemical treatment

Fine suspended particles in an effluent may be removed by coagulation and/or flocculation (Cooper, 1975; see also Chapter 10). Coagulation is essentially instantaneous whereas flocculation requires some more time and gentle agitation to achieve 'aggregation' of the particles. Ferrous or ferric sulphate, aluminium sulphate (alum), calcium hydroxide (lime) and polyelectrolytes are often used as chemical coagulants. A solution of coagulant of the appropriate strength for effective treatment is added to the effluent in a vigorously mixed tank, a precipitate or floc forms almost immediately and carries down the suspended solids to form a sludge. This sludge may be drawn off, mechanically dewatered and subjected to further treatment. The flocs formed on coagulation may be small, and will therefore require an extended period to settle, and as a consequence, for a given throughput of effluent a large sedimentation tank will be needed. Increasing the particle diameter by encouraging small flocs to coalesce (flocculation) increases the rate of sedimentation and thus, for a given throughput, a smaller vessel can be operated. Polyelectrolytes are commonly used as flocculants, and following addition the effluent is gently mixed (turbulent mixing would break up the flocs) by passage through sinuous flocculation channels, hydrodynamic flocculators or mechanically mixed flocculators (Smethurst, 1988).

Biological treatment

Most organic-waste materials may be degraded biologically. This process may be achieved aerobically or anaerobically in a number of ways. The most widely used aerobic processes are trickling filters, rotating disc contactors, activated sludge processes and their modifications. The anaerobic processes (digestion, filtration and sludge blankets) are used both in the treatment of specific wastewaters and in sludge conditioning.

Aerobic processes

TRICKLING FILTERS

The term filter in this unit operation is a misnomer, as the action of a trickling filter is not one of filtration, but rather it is a fixed film bio-reactor. Settled effluent to be treated is passed down through a packed bed counter-current to a flow of air. Micro-organisms ad-

hering to the packing matrix adsorb oxygen from the upflowing air and organic matter from the downflowing effluent; the latter is then metabolized and the effluent stream's BOD reduced.

A conventional trickling filter (Fig. 11.2) usually consists of a cylindrical concrete tank 2 to 3 m in depth and 8 to 16 m in diameter. Some filters are rectangular in shape, but a rotary system allows more uniform hydraulic loading (Bruce and Hawkes, 1983; Viessman and Hammer, 1993). The tank is packed with a bed of stone (usually granite) or special plastic packings, the bed being underlaid with drains. The packing material diameter should be 50 to 100 mm to give a specific surface of around $100 \text{ m}^2 \text{ m}^{-3}$, and the material should be packed to give a voidage (% air space of total bed volume) of 45 to 55%, which should minimize the risk of the spaces between the packing material becoming blocked by the microbial film. Synthetic packing material, although more expensive, has a higher surface area and voidage, allowing higher treatment rates per unit volume of bed and reducing the likelihood of blockages. The trickling filter is always followed by a secondary sedimentation tank or humus tank to remove suspended matter (e.g. biofilm sloughed off the packing) from the treated effluent. In conventionally loaded or low-rate filters, the effluent from which the suspended solids have been removed, is fed on to the upper surface of the bed by spray nozzles or mechanical distributor arms (McKinney, 1962; Higgins, 1968). The effluent trickles gradually through the bed and a slime layer of biologically active material (bacteria, fungi, algae, protozoa and nematodes) forms on the surface of the support material. The large surface area created in the bed permits close contact between air flowing upwards through the bed, the descending effluent and the biologically active growth. The bacteria in the biological film remove the majority of the organic loading. Complex organic materials are broken down and utilized, nitrogenous matter and ammonia are oxidized to nitrates and sulphides and other compounds are similarly oxidized. The higher organisms (protozoa etc.) control the accumulation of the biological film (prevents the filter from blocking) and improve the settling characteristics of the solids (humus) discharged with the filter effluent. In low-rate filters the scouring action of the hydraulic load normally has a minor role in removal of any loose microbial film. The active slime takes time to develop and can be poisoned by the addition of toxic chemicals. The simple filter is inefficient when operated at abnormally high organic loading rates. Initially, there is a very rapid build up of

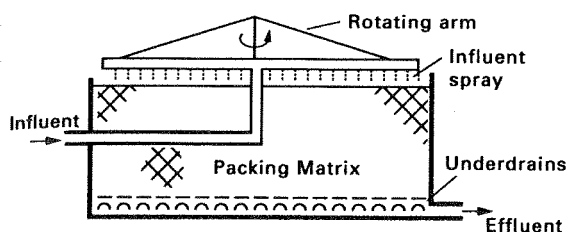


FIG. 11.2. Schematic diagram of a trickling filter.

bacteria, fungi and algae at the top of the filter which cannot be controlled by the resident population of worms and larvae. The voids, therefore, block up, resulting in ponding (untreated effluent accumulating on the surface of the filter bed). Film growth can be limited by reducing the dosing frequency, which gives better liquid distribution deeper into the bed.

A trickling filter bed should remove 75 to 95% of the BOD and 90 to 95% of the suspended solids at organic loading rates of $0.06\text{--}0.12\text{ kg BOD m}^{-3}\text{ day}^{-1}$ for conventional trickling filtration. When part of the treated effluent is being recirculated to dilute the feed and increase the hydraulic load placed on the unit the organic loading can be increased to $0.9\text{--}0.15\text{ kg BOD m}^{-3}\text{ day}^{-1}$. The increase in hydraulic load thus applied causes greater hydraulic scouring of the bed (preventing blockage), but does not reduce treatment efficiency due to improved wetting of the packing surface and, thus, more efficient use of the biofilm (Forster, 1985). To achieve the Royal Commission (20:30) Standard together with a high degree of nitrification, filters being supplied with domestic sewage should receive organic loading rates of $0.07\text{--}0.1\text{ kg BOD m}^{-3}\text{ day}^{-1}$ and hydraulic loading rates of $0.12\text{--}0.6\text{ m}^3\text{ m}^{-3}\text{ day}^{-1}$ (Gray, 1989).

It is possible to modify the trickling filter to increase the capacity for organic loading by the use of two sets of filters and settling tanks in series; this is known as alternating double filtration (ADF). Effluent is applied to the first filter at a high hydraulic and organic loading rate, it passes from this filter through the first settling tank and then on to a second filter and settler. After a period of one to two weeks the sequence of the filters is reversed and the second filter receives the higher loading. In this way heavy film growth is promoted in the first filter to receive the effluent, but when the filter sequence is reversed it becomes nutrient limited, encouraging excess film removal. Loading rates of $0.32\text{--}0.47\text{ kg BOD m}^{-3}\text{ day}^{-1}$ have been claimed (Forster, 1977), but recommended rates for design purposes are $0.15\text{--}0.26\text{ kg BOD m}^{-3}\text{ day}^{-1}$ (Forster, 1985).

Alternatively, enclosed deep beds of 3.5 to 5.5-m depth may be used in which air is blown through the beds by fans. Loading rates up to 12 times that of the ordinary filter have been claimed (Abson and Todhunter, 1967).

Cook (1978) has stressed the need to consider the possible intermittency of a factory wastewater treatment process. It was shown that starving a laboratory-scale trickling filter beyond 48 hours resulted in near failure of the filter. This indicated the need to supplement or artificially load the filter to maintain a viable biomass in the system.

TOWERS

Because trickling filters do not have both a high specific area and a high voidage, they are less suitable for the treatment of large volumes of strong industrial effluents (Table 11.2). Large areas of land would be required for the expensive and extensive traditional filter beds. Towers, 6–9 m in height, packed with lightweight ($40\text{--}80\text{ kg m}^{-3}$) plastic multi-faced modules or small random packing units have provided to be a space saving and relatively inexpensive solution to the problem. These packings have a relatively open structure for oxygen transfer (specific surface of $100\text{--}300\text{ m}^2\text{ m}^{-3}$) and high voidage (90–98%), but are expensive compared with the conventional filter packings. They are capable of coping with high BOD loadings. At a loading of $3.2\text{ kg BOD m}^{-3}\text{ day}^{-1}$ a 50% BOD removal may be achieved, and at $1.5\text{ kg m}^{-3}\text{ day}^{-1}$ 70% removal is possible (Ripley, 1979). The biological film is similar to that formed on the conventional packing and scouring is due to the hydraulic load applied rather than the predation of higher organisms.

BIOLOGICALLY AERATED FILTERS (BAFS)

Biologically aerated filters are a relatively recent development based on the trickling filter. They consist of a packed bed which provides sites for microbial growth through which air is passed but, unlike trickling filters, the reactor volume is flooded with the effluent to be treated which is passed upwards or downwards through the reactor (i.e. co- or counter-currently to the air supply) depending on the design. The packing matrix may be natural (e.g. pumice) or synthetic (e.g. polyethylene), and may be either a fixed structure or randomly packed.

The combination of aeration and filtration allows high rates of BOD and ammonia removal together with solids capture, so that sedimentation tanks may not be

required. However, regular backwashing is essential to remove filtered solids and excess biomass. Organic loading rates for 90% BOD removal are significantly greater than those obtained for trickling filters, being in the range $0.7\text{--}2.8 \text{ kg BOD m}^{-3} \text{ day}^{-1}$ (Stephenson *et al.*, 1993). They are versatile treatment systems, and of those currently in operation, design capacities vary between 600 and $70,000 \text{ m}^3 \text{ day}^{-1}$. In addition to their use as a secondary treatment process they can also be utilized for tertiary treatment or modified to allow denitrification in a manner similar to that of activated sludge systems.

ROTATING BIOLOGICAL CONTACTORS (ROTATING DISC CONTACTORS)

In this treatment method (Fig. 11.3) a unit composed of closely spaced discs (2 to 3 m diameter with 1 to 2 cm spacing between discs), on a central drive shaft are rotated slowly (0.5 to 15 rpm) through the effluent so that 40 to 50% of the disc surfaces are submerged (Borchardt, 1970; Pretorius, 1973). The discs, usually made from synthetic material (e.g. polystyrene, PVC), are arranged in stages or groups separated by baffles to minimize short circuiting or surging (Forster, 1985) and to enhance specific treatment requirements such as nitrification. The discs may be flat or corrugated to increase surface area. A microbial film forms on the discs; this is aerated during the exposed part of the cycle and absorbs nutrients during the submerged part. Shear forces produced as the discs rotate through the liquid control the thickness of the biofilm, with excess biofilm being sloughed from the discs. A sedimentation tank following the biological stage is therefore required to remove biological solids. Loading rates of $13 \text{ g BOD m}^{-2} \text{ day}^{-1}$ for domestic sewage and partial treatment of loads of $400 \text{ g BOD m}^{-2} \text{ day}^{-1}$ have been used. To achieve the 20:30 standard the loading rate should not exceed $6 \text{ g BOD m}^{-2} \text{ day}^{-1}$. Rotating biological contactors are compact, easily covered for health and aesthetic reasons, available as packaged units, simpler to operate under varying loads than trickling filters (the biofilm being wetted at all times) and are easily added onto existing treatment processes. As such they can provide a cost effective method of on-site treatment.

Ware and Pescod (1989) describe the use of full scale anaerobic/aerobic rotating biological contactors for treating brewery wastewaters. Greater than 85% COD removal was obtained in the aerobic stage, but difficulties were experienced in maintaining anaerobic populations.

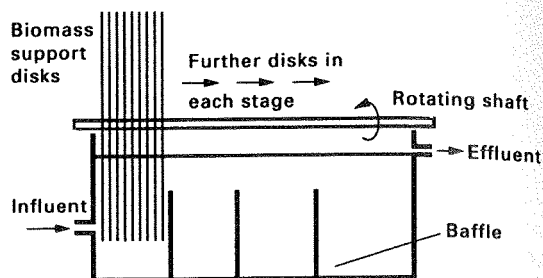


FIG. 11.3. Schematic diagram of a rotating biological contactor.

ROTATING DRUMS

Large rotating drums packed with random plastic packing materials or spheres have been manufactured as a development of rotating discs. Loading rates for the random packing were similar to those of rotating discs, while plastic spheres used in partial treatment could cope with loads of $6 \text{ kg BOD m}^{-2} \text{ day}^{-1}$ (Water Pollution Research Centre, 1972).

FLUIDIZED-BED SYSTEMS

Fluidized-bed reactors in wastewater treatment are relatively recent innovations. The support matrix (sand, anthracite, reticulated foam) has a large surface area on which the biofilm adheres and thus they are able to operate at high biomass concentrations with high rates of treatment. This allows strong wastewaters to be treated in small reactors. They are also useful for the treatment of industrial wastewaters when variable loadings are encountered (Cooper and Wheeldon, 1980, 1982). The support matrix is fluidized by the upflow of effluent through the reactor, and the degree of bed expansion is controlled by the flow rate of wastewater. The treated effluent can thus be decanted off without loss of the support matrix and with careful operation a secondary sedimentation tank may not be needed. The support matrix is regularly withdrawn to remove excess biomass. Fluidized-bed systems can be operated aerobically, anaerobically (see later section) or anoxically for denitrification.

ACTIVATED SLUDGE PROCESSES

The basic activated-sludge process (Fig. 11.4) consists of aerating and agitating the effluent in the presence of a flocculated suspension of micro-organisms on particulate organic matter — the activated sludge. This process was first reported by Arden and Lockett (1914)

and is now the most widely used biological treatment process for both domestic and industrial wastewaters. The raw effluent enters a primary sedimentation tank where coarse solids are removed. The partially clarified effluent passes to a second vessel, which can be of a variety of designs, into which air or oxygen is injected by bubble diffusers, paddles, stirrers, surface aerators etc. Vigorous agitation is used to ensure that the effluent and oxygen are in contact with the activated sludge. After a predetermined residence time of several hours, the effluent passes to a second sedimentation tank to remove the flocculated solids. Part of the sludge from the settlement tank is recycled to the aeration tank to maintain the biological activity. The overflow obtained from the settlement tank should be of a 20:30 standard or better and be suitable for discharge to inland waters. The excess sludge is dewatered and dried, to be sold as a fertilizer, incinerated or landfilled. In conventional activated sludge processes, organic loading rates are $0.5\text{--}1.5 \text{ kg BOD m}^{-3} \text{ day}^{-1}$ with hydraulic retention times of 5–14 hours depending on the nature of the wastewater, giving BOD reductions of 90–95%. High-rate activated-sludge processes can be used as a partial treatment for strong wastes prior to further treatment or discharge to a sewer and are widely used in the food processing and dairy industries. The organic loading rate is $1.5\text{--}3.5 \text{ kg BOD m}^{-3} \text{ day}^{-1}$, and with hydraulic retention times of only 1–2 hours, BOD reductions of 60–70% are possible (Gray, 1989).

A number of modifications of the basic process can be used to improve treatment efficiency, or for a more specific purpose such as denitrification (Winkler and Thomas, 1978; Gray, 1989). Tapered aeration and stepped feed aeration are used to balance oxygen demand (which is greatest at the point of wastewater entry to the aeration basin) with the amount of oxygen supplied. Contact stabilization exploits biosorption processes and thereby allows considerable reduction in

basin capacity ($\sim 50\%$) for a given wastewater throughput. Denitrification (the biological reduction of nitrate to nitrite and on to nitrogen gas under anoxic conditions) can be accomplished in an activated-sludge plant when the first part of the basin is not aerated.

In advanced activated-sludge systems the amount of dissolved oxygen available for biological activity is increased to improve treatment rate. One vessel of this type is the 'Deep Shaft' (Hemming *et al.*, 1977), which is quite distinctive from the other aeration tanks and has been developed from the ICI plc SCP process (Taylor and Senior, 1978; Chapter 7). The 'Deep Shaft' (Fig. 11.5) consists of a shaft 50 to 150 m deep, separated into a down-flow section (down-comer) and an up-flow section (riser). The shaft may be 0.5 to 10 m diameter, depending on capacity. Fresh effluent is fed in at the top of the 'Deep Shaft' and air is injected into the down-flow section at a sufficient depth to make the liquid circulate at $1 \text{ to } 2 \text{ m s}^{-1}$. The driving force for circulation is created by the difference in density (due to air bubble volume) between the riser and down-flow sections. For starting up, circulation of liquid is stimulated by injecting air at the same depth in the riser. Air injection is then gradually all transferred to the air injection point in the down-comer. Because of the pressure created in the down-comer, oxygen-transfer rates of $10 \text{ kg O}_2 \text{ m}^{-3} \text{ h}^{-1}$ can be achieved and bubble contact times of 3 to 5 minutes are possible instead of 15 seconds in diffused air systems. BOD removal rates of 90% are achievable at organic loadings of $3.7\text{--}6.6 \text{ kg BOD m}^{-3} \text{ day}^{-1}$ at hydraulic retention times of 1.17–1.75 hours (Gray, 1989). Sludge production was found to be much less than that for conventional sewage-treatment processes.

Two types of pure oxygen systems have also been developed to increase the rate of oxygen transfer:

- (i) closed systems which operate in oxygen-rich atmospheres and,
- (ii) open systems employing fine bubble diffusers.

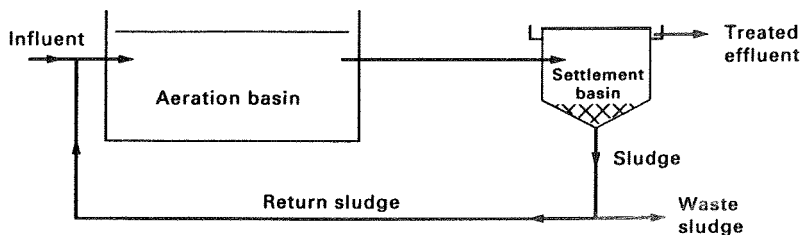


FIG. 11.4. Simplified cross-section of an activated sludge process.

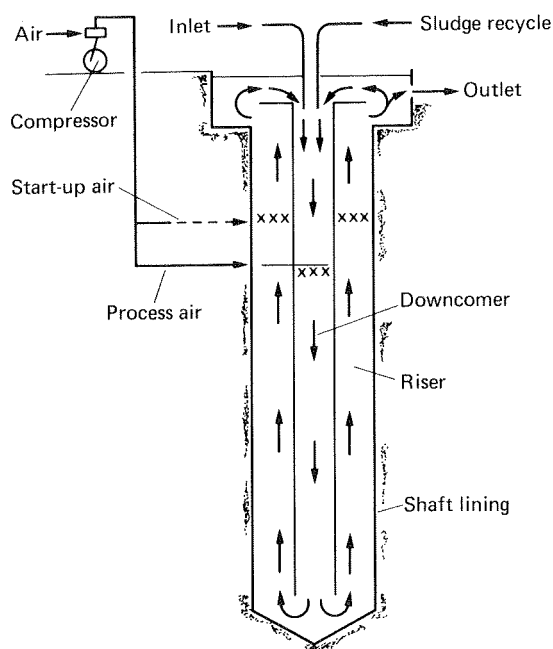


FIG. 11.5. Deep-shaft effluent treatment plant (Hemming *et al.*, 1977).

An example of the closed system is the UNOX Process developed by the Union Carbide Corporation in the U.S.A., and marketed in the U.K. by Wimpey Unox (Fig. 11.6). The enclosed oxygenation tank is compartmentalized by baffles. Settled wastewater and returned sludge are fed into the first stage and oxygen pumped into the headspace. The oxygen and wastewater move sequentially through the compartments, and the oxygen concentration in the gas phase decreases in each stage as it is consumed by the micro-organisms. As the nutrient concentration also falls stage by stage, oxygen supply and demand is balanced. Organic loading rates (when treating municipal wastewater) are 3–4 times higher than those of aerated systems at $2.5\text{--}4.0\text{ kg BOD m}^{-3}\text{ day}^{-1}$. Pure oxygen systems also have shorter residence times and produce less sludge with better settling qualities than conventional systems. The UNOX process has been successfully used for the treatment of brewery wastewaters containing $2000\text{ mg BOD dm}^{-3}$ at flow rates of $2269\text{ m}^3\text{ day}^{-1}$ with a 6.6 hour retention time (Brooking *et al.*, 1990).

The VITOX aeration system (Fig. 11.7) developed by the British Oxygen Company is an example of an open tank oxygenation system. Its main advantage is that it can be used in existing aeration tanks either to replace

or upgrade conventional aeration without the need to replace or modify existing units. Oxygenation is achieved by pumping settled wastewater at high pressure through a venturi where oxygen is injected. Turbulence and high pressures thus created ensure high levels of oxygen dissolution. The flexibility of this system means that it is particularly useful for the treatment of high strength intermittently produced wastewaters such as those generated by food processing industries (Gostick *et al.*, 1990).

Anaerobic treatment

Anaerobic treatment of waste organic materials originated with the use of septic tanks and Imhoff tanks, which have now been replaced by a variety of high-rate digesters (Pohland and Ghosh, 1971). Loehr (1968) has listed the following reasons for using anaerobic processes for waste treatment:

1. Higher loading rates can be achieved than are possible for aerobic treatment techniques.
2. Lower power requirements may be needed per unit of BOD treated.
3. Useful end-products such as digested sludge and/or combustible gases may be produced.
4. Organic matter is metabolized to a stable form.
5. There is an alteration of water-binding characteristics to permit rapid sludge dewatering.
6. The reduced amount of microbial biomass leads to easier handling of sludge.
7. Low levels of microbial growth will decrease the possible need for supplementary nutrients with nutritionally unbalanced wastes.

ANAEROBIC DIGESTION

Large volumes of wet sludge which are produced in primary and secondary sedimentation tanks may have to be reduced in volume before disposal. This volume of sludge can be reduced by anaerobic digestion. In sludges containing $20,000\text{--}60,000\text{ mg dm}^{-3}$ solid matter, 80% of the degradable matter may be digested, which will reduce the solids content by 50%. During anaerobic digestion acid fermenting bacteria degrade the waste to free volatile fatty acids, mainly acetic and propionic acid, which are then converted to methane ($\sim 60\%$) and carbon dioxide ($\sim 40\%$). The gas produced (biogas) is a very useful by-product, and can be burnt as a heating fuel, fed to gas engines to generate

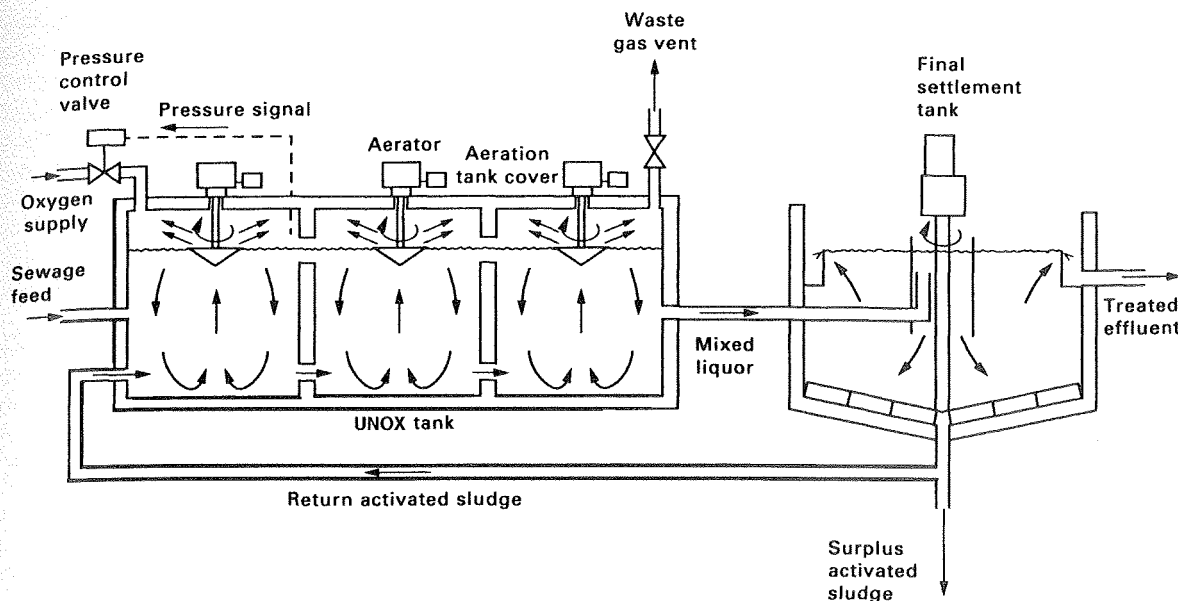


FIG. 11.6. Schematic diagram of a multistage UNOX system (Fuggle, 1983).

electricity or used as a vehicle fuel. As well as being used in sludge digestion and conditioning, anaerobic digesters are also used directly in the treatment of many high strength wastewaters, for example from the food and agricultural industries (Gray, 1989).

A number of anaerobic processes have been developed; completely mixed reactors (often these are simply described as anaerobic digesters) with or without sludge recycle, anaerobic filters, up-flow anaerobic sludge blankets (UASB) and anaerobic fluidized beds are the most common. Two stage anaerobic treatment systems utilizing a short retention time completely mixed acidification reactor followed by a UASB methanogenic reactor have been reported (Burgess and Morris, 1984; Ghosh *et al.*, 1985). Benefits claimed

include improved reliability, higher BOD/COD reductions and greater methane productivity.

ANAEROBIC DIGESTERS

The digester tanks used for this process may be up to 12,000-m³ volume and equipped with heating coils for accurate temperature control normally in the mesophilic range (25°–38°, usually 28°–32° in anaerobic digestion) to increase the rate of digestion, which is also improved by mechanical agitation (Loll, 1977). Some digesters are operated thermophilically (50°–70°, usually 55°–63° in anaerobic digestion) to increase the rate of degradation and biogas production, though much of the extra gas produced will be consumed in maintaining the digester temperature. The use of thermophilic anaerobic digestion has been investigated for the treatment of high strength (16–25 g COD dm⁻³), low pH (~3.8) and high temperature winery effluents (Romero *et al.*, 1988). Wiegant *et al.* (1985) reported the successful application of thermophilic anaerobic digestion of vinasse in completely mixed and UASB reactors, although methane production was found to decrease at high loading rates due to the presence of inhibitory compounds. Retention times are generally between 10 and 30 days, although with solids recycle retention times may be as low as 0.5 day. Most digesters operate on a semi-continuous basis with the

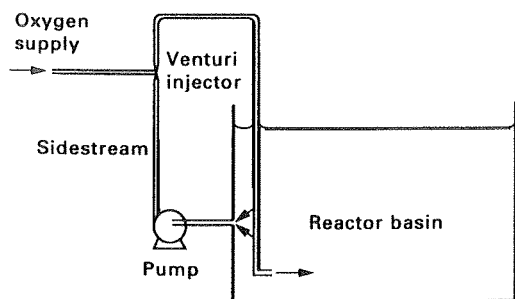


FIG. 11.7. Simplified VITOX sidestream aeration system.

appropriate volume of digested material being removed and replaced with 'fresh' material on a daily basis.

ANAEROBIC FILTERS

In anaerobic filters, as with the aerobic filters, there is a microbial film growing on an inert support (Chian and DeWalle, 1977). Anaerobic filters may be operated in an upflow or downflow mode, and a wide variety of packings are used; both natural and synthetic (Young, 1983). The first full-scale anaerobic filters were constructed in the 1970s and in 1982 the Bacardi Corporation brought into operation a 100,000-m³ plant to treat distillery effluents (Szendrey, 1983). Many other effluents are found to be amenable to treatment using anaerobic filter systems; antibiotic fermentation wastes, citric acid fermentation wastes, yeast production wastewater and brewery and winery wastewaters (Szendrey, 1983), molasses distillery slops (Silvero *et al.*, 1986) and fermentation and pharmaceutical wastes (Bonastre and Paris, 1989).

UP-FLOW ANAEROBIC SLUDGE BLANKETS (UASB)

In this system (originally developed in the Netherlands) high levels of active biomass are retained in the reactor by flocculation (Lettinga *et al.*, 1980, 1983). No support media is added to the reactor, instead the flocculated sludge develops in the reactor and acts as a fluidized bed. Feed is pumped through the bed (the sludge blanket), above which fine particles flocculate and settle back to the blanket as sludge, thus preventing washout of organisms. Anaerobic sludge blankets have been found to be effective in the treatment of many wastewaters, including sugar-beet wastes, domestic sewage, slaughterhouse wastes, agricultural wastes (Lettinga *et al.*, 1983), brewery wastes (Fang *et al.*, 1989, 1990), winery wastes (Cheng *et al.*, 1990) and distillery wastes (Burgess and Morris, 1984; Ghosh *et al.*, 1985).

BY-PRODUCTS

The marketing of wastes from fermentation processes has been established for at least 200 years. By the 1700s, brewers' grains, spent hops and surplus yeast from larger breweries were accumulating in sufficient quantities for specialized trades to develop (Mathias, 1959). Around London, cattle and pigs were fattened on 'wash' and brewers' grains. Excess yeast was supplied to bakers and gin distillers.

In any fermentation recovery process there is a need

to recognize whether there is a potential marketable waste and, if necessary, to develop a market. Obviously the marketability and cost of reclamation of by-products will be very important in deciding upon a policy of waste recovery. Under favourable market conditions it has been claimed that the profit on animal feed by-products from a completely integrated distillery may almost pay the cost of the grain (Blaine, 1965). This claim would now be more difficult to substantiate, due to fuel costs and capital outlay on plant (Quinn and Marchant, 1980; Sheenan and Greenfield, 1980).

Some microbial processes yield residues which are difficult or impossible to sell (Blaine, 1965). In these processes it may be possible, and worth while, to change to low residue raw materials such as refined sugars or other pure compounds as sources of soluble nutrients.

Distilleries

In grain-based distilleries it has been common practice to recover spent grain and stillage, the waste liquor after the alcohol has been distilled off (Boruff, 1953; Blaine, 1965). The stillage is first passed through screens with 1-mm openings. The screenings are then dewatered with mechanical filter presses and dried in rotary driers (Chapter 10). This product is termed Distillers' Dried Grains (light grains). The screened stillage is concentrated in evaporators to give 25 to 35% solids in a thick syrup. Stillages which have been prefiltered may be concentrated to 35 to 50% solids. This syrup can then be mixed with the pressed screenings and dried to give Distillers' Dried Grains with Solubles (dark grains). Alternatively the evaporated stillage may be dried completely in drum driers to produce Distillers' Dried Solubles. Dark and light grains and dried solubles have all been used as animal-feed supplements. Flachowsky *et al.* (1990) report their use following chemical treatment as a replacement feed for sheep. Dried solubles have also been used as a medium adjunct in the preparation of antibiotics (Chapter 4).

In distilleries using cane molasses as a feedstock, evaporated spent wash has been used as a fuel for boilers (Sheenan and Greenfield, 1980). It has proved worth while to recover potassium salts from sugar-beet stillage. The market for the evaporated product must be considered within a range of 50 km of the evaporation plant (Lewicki, 1978).

Boruff (1953) cited an unpublished 1949 survey of American distilleries which showed that 85% of the stillage solids were being recovered as dried feeds, 14% solids as wet grains and only 1% was waste. In recent

years the traditional distillers' by-products from stillage have become increasingly uneconomic. A number of processes to produce SCP from stillage using *Geotrichum candidum*, *Candida utilis* or *C. tropicalis* have been evaluated (Quinn and Marchant, 1980; Sheenan and Greenfield, 1980).

Breweries

The three marketable wastes from breweries are spent grain, spent hops and yeast. The spent grain is recovered from the mash tun and is then sold as animal feed either after pressing in a wet state or after drying in rotary driers. Alternatively, the wet grain may be used in the preparation of silage for cattle. The possible markets for hops are restricted. Some are used as a fertilizer or as a low-grade fuel.

Yeast can be separated from beer by filtration or centrifuging. The yeast slurry is then dried in drum driers. Some of the yeast may be mixed with brewers' spent grains to produce a feed material with a slightly higher protein content than normal brewers' grains. The yeast may also be used directly as a source of vitamins. If it is to be used as a human food it must be debittered to remove the hop bitter substances absorbed on to the yeast cells. The cells are then washed in an alkaline solution, washed with water and drum dried. Although bakers' yeast was originally obtained as a brewery by-product, this market has diminished considerably. Most bakers' yeast is now produced directly by a distinct production process. Dewatered sludge from brewery wastewater treatment operations has been reported to increase agricultural yields when used as fertilizer (Naylor and Severson, 1984). Lyons (1983) reports on the potential use of brewery effluents as a feedstock for fuel and industrial ethanol production.

Amino acid wastes

The main wastes from glutamic acid or lysine fermentations are cells, a liquor with a high amino-acid content which can be used as an animal-feed supplement, and the salts removed from the liquor by crystallization, which is a good fertilizer (Renaud, 1980).

Fuel alcohol wastes

The stillage from ethanol production and wastes from starchy fermentations are, following concentration, saleable as an animal-feed supplement. Wastes

from sugar fermentations and distillation can be digested anaerobically and the methane generated used as an energy source (Essien and Pyle, 1983; Faust *et al.*, 1983; Singh *et al.*, 1983). Faust *et al.* (1983) also suggest the use of CO₂ rich off gases in the food and beverage industries.

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Fermentation Economics

INTRODUCTION

IF A FERMENTATION process is to yield a product at a competitive price, the chosen micro-organism or animal cell culture should give the desired end-product in predictable, and economically adequate, quantities. A number of basic objectives are commonly used in developing a successful process which will be economically viable.

1. The capital investment in the fermenter and ancillary equipment should be confined to a minimum, provided that the equipment is reliable and may be used in a range of fermentation processes.
2. Raw materials should be as cheap as possible and utilized efficiently. A search for possible alternative materials might be made, even when a process is operational.
3. The highest-yielding strain of micro-organism or animal cell culture should be used.
4. There should be a saving in labour whenever possible and automation should be used where it is feasible.
5. When a batch process is operated, the growth cycle should be as short as possible to obtain the highest yield of product and allow for maximum utilization of equipment. To achieve this objective it may be possible to use fed-batch culture (Chapters 2 and 4).
6. Recovery and purification procedures should be as simple and rapid as possible.
7. The effluent discharge should be kept to a minimum.
8. Heat and power should be used efficiently.

9. Space requirements should be kept to a minimum, but there should be some allowance for potential expansion in production capacity.
10. All the above must comply with safety guidelines and regulations.

The consideration of so many criteria means that there may have to be a compromise for the particular set of circumstances relating to an individual process. Winkler (1991) identified three key economic objectives in the fermentation stage: maximum product yield, process productivity and substrate utilization. However, these criteria may be overridden by the demands of subsequent purification stages for high product concentration and high product purity which may in turn be overridden by safety considerations.

In any process it is important to know the cost breakdown (Table 12.1), so that it may be seen where the biggest potential savings may be achieved. In a review of a number of processes, Nyiri and Charles (1977) concluded that four basic components contributed to the process cost in the following decreasing order: raw materials, fixed costs, utilities, labour. When raw materials are a major part of the total cost it is obvious that media and microbial strain-improvement research should form a major part of a development programme. Development work on components contributing little to the cost of the product could not be justified.

Costs quoted in this chapter have been taken from the source of information and are applicable only at the time of that publication. Atkinson and Mavituna (1991) have discussed the use of cost indices to update historical data.

Caution is necessary when examining some examples of potential fermentation processes, since the capital and operating costs normally have been estimated us-

TABLE 12.1. Production costs breakdown expressed as percentages of total cost

Item	Product							
	Beer (Pratten, 1971)	Alcohol (batch) (Maiorella, 1984)	Acetic acid (Pape, 1977)	Citric acid (Schierholt, 1977)	Norprotein (Mogren, 1979)	SCP (Anon, 1974)	Penicillin (Swartz 1979)	tPA (Datar <i>et al.</i> 1993)
Raw materials	38.4	76.7	42.2	39.7	70.0	62	58.0	39.8
Utilities	*	11.7	23.1	35.3	16.0	10	20.3	20.5
Labour and supervision	24.5	2.9	19.5	25.0	9.0	9	5.4	10.9
Fixed charges	7.2	4.8	10.5	—	—	19	—	—
Maintenance	29.9	—	—	—	5.0	†	14.9	—
Operating supplies	—	—	—	—	—	—	1.4‡	—
Waste	—	—	—	—	—	—	—	12.0
Materials	—	—	—	—	—	—	—	13.3
recovery	—	—	—	—	—	—	—	—
Other	—	—	—	—	—	—	—	3.6

Note * Included in 29.9% for maintenance and operating supplies.

† Included in 9% for labour and supervision. ‡ 0.2% for laboratory costs included.

ing chemical engineering costing principles. Accurate detailed costing of industrial fermentations are rarely published.

Government aid or taxation can determine the viability of many fermentation processes. Support in construction of plant, development and production programmes of acetone-butanol and penicillin during wartime led to production of compounds at an earlier stage than might normally be regarded as economically feasible (Hastings, 1971, 1978). Agricultural-aid programmes in the United States of America made available low-cost supplies of grain and potatoes and enabled fermentations to be operated when they would not have been economically viable in a free market (Perlman, 1970). In 1980 carbohydrates as molasses or cane juice could be obtained in sugar-producing countries at half the price of molasses in the European Economic Community (Meers, 1980) and differences in starch prices between the EEC and the U.S.A. were about 40% (Gray, 1987). This price differential of carbon substrates discouraged investment and further research in the EEC. Policy changes were made to allow fermentation companies within the EEC to buy sugar and starch based substrates at a lower price to enable them to compete on a world-wide scale.

Some very useful reviews of process economics have been produced for penicillin G (Swartz, 1979), gibberellic acid (Vass and Jefferys, 1979), biomass from natural gas (Hamer, 1979), biomass from whey (Meyrath and Beyer, 1979), biomass from waste carbohydrates (Mateles, 1975), biomass from cane and coffee process-

ing by-products (Rolz, 1975), 6-aminopenicillanic acid (Harrison and Gibson, 1984), evaluation of 11 alternative ethanol fermentation processes (Maiorella *et al.*, 1984), tissue plasminogen activator (Datar *et al.*, 1993), primary separation steps (Datar, 1986) and chromatography (Sofer and Nystrom, 1989). A number of general reviews of fermentation economics are also available (Whitaker, 1973; Nyiri and Charles, 1977; Bartholomew and Reisman, 1979; Stowell and Bateson, 1984; Bailey and Ollis, 1986; Hacking, 1986; Kalk and Langlykke, 1986; Reisman, 1988; Atkinson and Mavituna, 1991).

ISOLATION OF MICRO-ORGANISMS OF POTENTIAL INDUSTRIAL INTEREST

The most appropriate micro-organism for a potential process is usually found by isolation from a variety of sources, most commonly soil. The classical method of screening to obtain a suitable organism tended to be very time consuming, expensive and often without a very clear objective. Eli Lilly and Company Ltd. discovered three new antibiotics in 10 years while screening 400,000 micro-organisms (Nelson, 1961). More recently Berdy (1989) has speculated that the screening of 100,000 soil micro-organisms may lead to the isolation of 5 to 50 new compounds, but there is no guarantee after evaluation that a useful new drug or other product will be found.

If a desired characteristic, which gives the organism a selective advantage, has already been recognized, a

screen might be designed incorporating this characteristic as a selective factor (Chapter 3). The isolation may begin with pretreatment of samples which favour the survival of the preferred organism. This is followed by growth on selective or non-selective media and often associated with batch or continuous enrichment. Important factors which will be of economic significance which might be selected in a well planned screen could include:

1. Growth on a simple cheap medium.
2. Growth at a higher temperature (to reduce cooling costs).
3. Better resistance to contamination.

However, the synthesis of other microbial products (e.g. antibiotics) does not give the producing organism any selective advantage which might be used in an isolation procedure (Chapter 3). Therefore a collection of these organisms must be made before testing for the desired characteristic. Because many isolation procedures will lead to the rediscovery of known organisms with known activities it is important to use well planned, efficient isolation procedures which can prove very productive.

A number of approaches are currently being used to improve isolation procedures (see also Chapter 3). Numerical taxonomic data bases are being exploited to design selective media for certain microbial taxa. Using these data bases it has been possible to design media to encourage the growth of uncommon streptomycetes or discourage the growth of common species (Vickers *et al.*, 1984; Williams and Vickers, 1988). Knowledge of antibiotic sensitivity gained from taxonomic data bases has led to the design of other selective media which will select for resistant groups of organisms (Goodfellow and O'Donnell, 1989; Bull, 1992; Bull *et al.*, 1992). The selection of antibiotic producing soil isolates has also been achieved using media designed by a stepwise discrimination analysis technique (Huck *et al.*, 1991).

These 'designed' isolation media are now being used extensively for the isolation of novel and rare microorganisms. For example, within the actinomycetes, *Streptomyces* spp. have been extensively screened since the 1940s for antibiotic production with subsequent notable commercial success. More interest has now been shown in isolation of strains of the less common genera such as *Actinomadura*, *Actinoplanes*, *Kitosatosporia*, *Streptoalloteichus*, etc. which are producing other novel bioactive compounds (Goodfellow and O'Donnell, 1989; Bull, 1992).

It has become a common practice to obtain isolates from unusual habitats, which may include extreme environments, to ensure that the greatest microbial diversity is being examined (Bull, 1992; Bull *et al.*, 1992; Chapter 3).

The screening tests which have been developed to detect new useful compounds of potential industrial interest have become much more selective and sensitive. Better knowledge of cell biochemistry has enabled the design of screens which are much more precisely targeted to detect the desired activity using specific detector strains (Chapter 3). Many large companies which undertake screening programmes have introduced some automation to enable high throughput rates which are cost effective and less labour intensive. In 1989, Nisbet and Porter considered that 10^5 tests per year with up to 20 assay tests should be undertaken in a worthwhile screening programme.

STRAIN IMPROVEMENT

Strain improvement using a mutation/selection programme (Chapter 3) for improving an organism being used in an established process or a potential process can be very cost effective. Historically, mutation/selection programmes to improve strains of *Penicillium chrysogenum* were time consuming, labour intensive and very random because of the lack of knowledge about penicillin biosynthesis. These mutation programmes did, however, contribute significantly to increases in penicillin yields from less than 100 units cm^{-3} in the 1940s to over 51,000 units cm^{-3} by 1976 (Queener and Swartz, 1979) and a four-fold increase in yields between 1970 and 1985 at Gist Brocades (Royce, 1993). Improvements for streptomycin, chlortetracycline and erythromycin are reported in Table 3.8.

It is always very important to decide if a strain improvement programme can be justified on financial grounds to improve the overall economy of a process. Lockwood and Streets (1966) quoted the example of a fermentation process making 453,600 kg year^{-1} of product at 23.5 p kg^{-1} in which a 1% increase in the product would produce an increased return of only £1070, which they considered insufficient to support a worthwhile mutation programme. If the output had been 10 times greater, a 1% increase would have produced £10,700 which was thought to be just sufficient to meet the costs of research. If this output could have been increased to 10%, the increased return would have been £107,000, which would have been much larger than the cost of a mutation programme at

that time. Calam (1969) suggested that a graduate worker with two assistants are capable of operating one or two mutation research programmes and of making effective progress.

A better understanding of cell metabolism and its regulation has enabled the development of more logical targeted methods to be introduced to select for mutants with desirable 'blue prints' where there may be a need to block undesirable enzyme activities or eliminate negatively acting control mechanisms (Chapter 3). This approach is much more efficient and economic in terms of resources and time. It was first employed extensively in the preparation of mutant strains used in amino acid fermentations.

Although the main targets in strain improvement are normally to increase the product yield or specific production rates, it is also important to consider strain stability, resistance to phage infection, response to dissolved oxygen, tolerance to medium components, production of foam and the morphological form of the organism. Methods to achieve many of these changes are discussed in Chapter 3. These are very important in helping to achieve targets in a research and development programme as they can have a significant impact on the process and/or product (Table 12.2). A study of the range of targets given in this table, which might be aided by strain improvement, has economic effects on all aspects of a fermentation process.

There are a number of companies with special expertise, such as Cetus Ltd and Panlabs Inc., who will perform strain-development programmes or make cultures available for commercial clients. An example of work by Panlabs Inc.'s improvements in yields of penicillin from cultures of *P. chrysogenum* during 1973 to 1976 is shown in Table 12.3 (Queener and Swartz, 1979). These data make it possible to compare yields from the same culture cultured at various scales.

MARKET POTENTIAL

The fermentation technologist should be aware of the problem of assessing market potential, although he/she may not be primarily involved in collating or assessing the necessary data. Some aspects have been considered by MacLennan (1976), Hepner (1977), Lawson and Sutherland (1978) and Keim and Venkatasubramanian (1989).

Four categories of microbial product can be recognized economically and it is important to consider to which category a compound belongs:

1. Low price bulk chemicals, e.g. solvents, biomass, high fructose syrups ($\text{US\$}10^2\text{--}10^3 \text{ tonne}^{-1}$).
2. Mid price chemicals, e.g. organic acids, amino acids, bipolymers ($\$10^3\text{--}10^5 \text{ tonne}^{-1}$).
3. High price microbial and animal-cell products, e.g. enzymes, vitamins, antibiotics, corticosteroids, vaccines, etc. ($\$10^5\text{--}10^7 \text{ tonne}^{-1}$).
4. Very high-price animal-cell products, e.g. monoclonals, tissue plasminogen activator, etc. ($\$10^7\text{--}10^9 \text{ tonne}^{-1}$).

The third and fourth groups can normally be produced only by a microbial or animal-cell based process and therefore do not have to compete with an alternative chemical process which is usually much cheaper. This includes compounds which have complicated structures, are chemically or thermally unstable or for which a multi-stage chemical synthesis would be expensive. Many microbial products are not exploited because cheaper synthetic processes are available.

Hepner (1977, 1978) has examined the factors that determine the feasibility of large-scale ethanol production by fermentation. He considered that ethanol produced by fermentation would only be competitive with synthetic ethanol from crude oil if the fermentation plant was in an area where cheap supplies of carbohydrate were available. In an example based on 1977 costs, if crude oil cost $\$100 \text{ tonne}^{-1}$, fermentation produced ethanol would be financially viable only if raw-sugar feedstock cost less than $\$109 \text{ tonne}^{-1}$ or molasses cost less than $\$75 \text{ tonne}^{-1}$ (Hepner, 1978). In Brazil in 1975, an ethanol production programme using sugar-cane as a substrate was started so that the petroleum imports could be reduced. During 1977, $7 \times 10^8 \text{ dm}^3$ of ethanol were produced and it was hoped to have increased production to $1.5 \times 10^9 \text{ dm}^3$ during 1979. The ethanol, although subsidized by the Brazilian government, was selling at $\$1$ per 4.5 dm^3 , which was more than the cost of petrol refined from imported oil (Hammond, 1978). Other aspects of potential ethanol production have been discussed by Bu'Lock (1979).

It is necessary to estimate the size of the present and potential market and the increase in demand for a compound. This type of exercise was undertaken for single-cell protein (MacLennan, 1976; Taylor and Senior, 1978). Taylor and Senior (1978) gave summaries of major single-cell protein plants which were operational or planned, estimates of single-cell protein production in 1980 and 1985 and predicted world supply and demand for high-quality protein meal. It was estimated that by 1985 there would be a market of 5×10^6

TABLE 12.2. *Criteria for strain improvement* (Schwab, 1988)

Target	Impact on process or product
Improvement of titre and/or specific production rate	General decrease of production costs, improved exploitation of reactor capacity, lower investment costs, increased efficiency in downstream processing steps
Improvement of yield	Lower costs for substrates, decreased production of heat and CO ₂ , lower cooling costs, less waste and pollution
Change in catabolic capabilities	Use of more favourable substrates (less expensive, better availability, etc.), omission of pretreatment steps (e.g. enzymatic hydrolysis of polysaccharides).
Improvement of technological features of micro-organisms (e.g. flocculation behaviour, structure of mycelium, sporulation, foaming, strain stability, etc.)	Less energy costs for mixing and oxygen transfer, improved separation characteristics, fewer problems in inoculum preparation or scale-up of the process
Improvement of product quality	Decreased production of specific by-products (fewer impurities), prevention of product degradation (e.g. pectinases)
Modification of products	Improvement of solubility in extraction solvents (e.g. addition of specific side chains), increased thermic stability of altered enzymatic properties of proteins
Changing of the locus of product accumulation (e.g. intracellular to extracellular)	Improved product recovery (e.g. omission of cell disruption), correct products (e.g. fully processed proteins).

tonnes annum⁻¹ for single-cell protein world wide, whereas production would only be 2.9×10^6 tonnes annum⁻¹. Such predictions supported the establishment of a process by ICI plc. Unfortunately the product which was marketed as an animal protein feed during the 1980s could not compete for price with soya beans and the manufacturing plant was closed down (Sharp, 1989).

Ratafia (1987) estimated the world markets in 1991

for products that can be made by mammalian cell culture to be as follows ($\$ \times 10^6$): diagnostics 6500, vaccines 5200, hormones 4900, lymphokines 1750, monoclonal antibodies 1700, other products 3040. The largest market for a cell culture product was indentified for tissue plasminogen activator (tPA) which was being used in clinical trials and proving effective in dissolving several forms of blood clots. Datar *et al.* (1993) have estimated the potential number of patients in the

TABLE 12.3 Panlab's penicillin strain-development programme, P-line, showing yields in units cm^{-3} from data supplied by Panlabs Inc. (Queener and Swartz, 1979)

Year	Culture	Penicillin	Shaker yield			Time (days)	Pilot-plant yield			Time (h)	Production yield			Time (h)
			Min.	Max.	Av.		Min.	Max.	Av.		Min.	Max.	Av.	
1973	P-1	V	6,100	7,500	7,100	5	14,500	17,600	15,900	140	11,700	14,400	13,560	130
	P-3	V	11,000	13,100	12,000	8	13,900	20,700	18,000	180	14,200	19,100	16,500	180
		V	17,400	21,300	19,800	9	15,400	21,800	18,200	182				
		G									17,600	17,900	17,700	132
	P-4	V	23,300	25,400	24,400	10			22,100	182	29,700	20,700	20,120	182
1974		G									15,800	20,600	17,900	182
	P-5	V	27,500	31,200	29,800	10	22,600	23,700	23,150	182	23,600	25,900	24,700	182
		G									21,300	24,600	23,150	182
	P-7	V	29,200	33,900	32,100	10	25,100	33,400	30,200	210	23,400	29,600	27,300	190
		G					25,200	29,000	27,700	185	23,700	28,200	25,600	190
1975	P-8	V	34,700	36,900	36,000	10	31,500	33,000	32,250	185	29,200	34,800	32,400	203
		G									27,900	30,800	29,600	203
	P-10	V	36,800	42,600	38,700									
1976	P-10	V					32,500	36,500	33,700	185				
	P-12	V	40,911	42,800	41,900	9	32,300	38,000	35,000	185				
	P-13	V	33,700	39,000	37,200	7			41,500	185				
		G							42,250	185				

U.S.A., Europe and Japan who would benefit from treatment with tPA to be 2.8×10^6 (about 0.58% of the human population). The price of the FDA licensed tPA is currently \$2,200 per dose or \$22,000 g^{-1} .

At this stage it is worthwhile comparing actual sales of recombinant products with those of some other fermentation products. Since their introduction in the 1980s, the total sales of four new secondary metabolites, cyclosporin (an immunoregulator), imipen (a broad spectrum antibiotic), lovastatin (controller of cholesterol levels) and ivermectin (antiparasitic compound), have been higher than those of all the recombinant products (Buckland, 1992).

The life expectancy of a compound will have to be predicted even when covered by a patent. This is sometimes difficult, as has been demonstrated with the industrial enzyme market (Aunstrup, 1977). In about 1965 detergent enzymes became widely used, which led to a general increase in sales of microbial enzymes. In 1971 allergic symptoms were discovered in workers handling enzymes in a detergent factory, resulting in the removal of enzymes from most detergent powders and a sudden drop in enzyme sales. Enzyme sales recovered after the introduction of improved process techniques.

SOME EFFECTS OF LEGISLATION ON PRODUCTION OF ANTIBIOTICS AND RECOMBINANT PROTEINS IN THE U.S.A.

There are a number of differences in regulatory

requirements for production facilities for antibiotics and recombinant proteins (Bader, 1992). In both cases, the products must be licensed by the FDA before they can be marketed.

The recombinant protein is produced by a precisely specified process using high quality substrates and processed and purified in an aseptic pharmaceutical facility that has been inspected and licensed by the CBER (Center of Biological Evaluation and Research) of the FDA. Steps must also be taken to ensure that there is no cross contamination if more than one product is produced in the same facility. Any changes in a process or facilities must be approved by the CBER before implementation. It may take 7 years to obtain approval by the CBER for a production plant. In contrast, the requirements for an antibiotic production plant are less stringent and it may be operational in 4 years. The delay in start up to produce the recombinant protein will result in much higher costs.

It is important to remember that the detailed clinical evaluation of a microbial compound as a drug, plus FDA approval, may take 8–10 years from initial discovery and cost up to $\$150 \times 10^6$.

PLANT AND EQUIPMENT

It is most logical to build equipment as large as possible because of the economy of scale. There is an empirical relationship between cost and size of an item

of equipment. According to this relationship, as facility size increases, its cost increases thus:

$$\frac{\text{cost}_1}{\text{cost}_2} = \left(\frac{\text{size}_1}{\text{size}_2} \right)^n$$

where n is an exponent or scale factor. Scale factors have been estimated to be 0.6 for brewing (Pratten, 1971), 0.7 to 0.8 for a single cell protein plant (Humphrey, 1975; MacLennan, 1976), 0.6 for antibody production (Birch *et al.*, 1987), 0.75 for fermentation processes (Bartholomew and Reisman, 1979) and 0.6 for waste water treatment (Eckenfelder, 1989).

In brewing, when cylindrico-conical vessels are used, there would appear to be no economic advantage in scaling up above 108,000-dm³ capacity (Hoggan, 1977), although vessels of 360,000 dm³ have been installed. In some breweries a number of smaller vessels have been installed to allow for fluctuating demand of different beers.

The operational vessel volume is a critical factor when considering high volume-low cost products. In processes carried out at volumes greater than 100 m³, the use of an air-lift fermenter is more economical as the relative investment costs per unit of output decreased more rapidly than for a stirred type of fermenter (Table 12.4).

However, a number of restraints have to be considered before deciding on the scale of operation. Such restrictions include cooling and aeration requirements and the method of fermentation vessel construction. The need for cooling provisions in many fermentations has already been described (Chapter 7). At this stage it is important to remember that the volume of a fermenter is proportional to r^3 (where r is the fermenter radius), whereas the increase in surface area is proportional to r^2 . Therefore the scaling up of a vessel will lead to a decrease in the surface area to volume ratio and therefore a decrease in the effectiveness of a cooling jacket. There may be a fermenter volume above

which it will not be possible to remove enough heat to maintain constant temperature, unless the cooling capacity is increased by incorporating internal cooling coils or by using external heat exchangers. These modifications may prove costly or else interfere with mixing in the vessels. There is possibly the alternative of using micro-organisms with higher optimum temperatures. The oxygen requirements of a process may limit the size of vessel which can be operated successfully. In acetic acid (vinegar) production, where very efficient aeration is critical, because *Acetobacter* spp. are very sensitive to oxygen depletion in the broth, the maximum acetator capacity which could be used was about 50,000 dm³ (Pape, 1977).

There is an upper size limit for a custom-built fermenter that can be transported to a site, while much larger vessels may be built on site. Most countries limit the maximum size of unit that can be transported on the road. In 1979 the first ICI plc production vessel for single-cell protein was installed; this had a volume of 1.5×10^6 dm³, was 80-m tall and cost $\text{£}6 \times 10^6$ (Sharp, 1989). It was constructed in France, floated on a barge to the River Tees, and transported a very short distance on land. At the time of erection it was the largest fabricated fermenter ever to be transported.

Because of the capital investment and operational costs there is now a trend to consider unconventional fermenter designs of simple construction with very efficient oxygen-transfer to be used for specific purposes, particularly single-cell protein. In this context, Schugerl *et al.* (1978) have considered the case of a single-cell protein plant for 100,000 tonnes annum⁻¹ using methanol as the main substrate. The plant investment was only 20% of the production cost. In this investment cost only 20 to 25% was due to the fermenter. Of this, the vessel accounted for 5 to 10%, the stirrer and aerator 5 to 10% and cooling 10 to 15%. Energy, water, aeration and auxiliaries were estimated to be a further 10% of the production costs. Since the fermenter investment costs are approximately 4 to 5% and the operating costs are 10%, it was concluded that the type of vessel could influence only 15% of the total production costs. If a tower fermenter were used, the main advantage would appear to be lower operating costs, particularly reduced energy and cooling-water costs, as the removal of mechanical stirring would diminish the cooling water requirements.

A useful guide has been prepared by the Institution of Chemical Engineers (1977) which outlined, in reasonably quantitative terms, many factors making up a check list which must be incorporated into the final

TABLE 12.4. Estimate of relative costs of a fermentation process as a function of scale (100 m³ = 1 unit of volume / unit of time) (Van Suydam, 1987).

Scale (m ³)	Cost scale factor
1000	0.3
200	0.6
100	1.0
10	10.0
1	30.0

capital costing of a chemical-plant project. It is possible from a knowledge of the proposed plant location, a sketch of the chemical plant process flow sheet, the size of the major items of equipment and the service requirements, to estimate capital and operating costs to $\pm 15\%$ (Backhurst and Harker, 1973). Unfortunately, comparable literature for fermentation plants is not available.

Some details of the cost of equipment used in fermentation processes has been discussed by Whitaker (1973), Humphrey (1975), Mateles (1975), Rolz (1975), Nyiri and Charles (1977), Maiorella *et al.* (1984), Hackling (1986), Kalk and Langlykke (1986), Reisman (1988) and Atkinson and Mavituna (1991). It is essential to remember that most of the data are estimates for proposed processes. Some equipment-cost breakdowns are quoted in Table 12.5.

Kalk and Landlykke (1986) quote 1985 costs of fer-

menters. In the unit size range of 1.0 to 45 m³, costs for modular units with control and recording ranged from \$90,000 to \$350,000. The cost of 70 to 250 m³ vessels, which needed site erection, ranged from \$800,000 to \$2,000,000.

If a vessel is to be used to produce genetically engineered products, the costs will increase significantly because of the extra containment provisions which must be incorporated during construction (see also Chapter 7). It has been estimated that the basic fermenter costs will increase by 10–30% for each increase in Containment level (Hambleton *et al.*, 1991). At level 2 or B3 (see Chapter 7 for definition) the provision of containment of an instrumented 20-dm³ fermenter may increase the cost from £45,000 to £100,000. The cheaper alternative might be to enclose an ordinary fermenter in a suitable containment cabinet. Hambleton *et al.* (1991) have reported the use of a

TABLE 12.5. Capital cost breakdown for fermentation plant

Item	% of total
(a) <i>Penicillin plant</i> , estimated for five 225,000 dm ³ fermenters with ancillary equipment (Swartz, 1979)	
Process equipment	23.6
Installation	5.2
Insulation	1.9
Instruments	2.7
Piping	11.8
Electrical	15.8
Building	11.3
Utilities	21.3
Site	2.4
Laboratory equipment	3.8
Spare parts	0.5
(b) <i>Norprotein plant</i> (Mogren, 1979)	
Raw materials storage	10
Media preparation and utilities	17
Fermentation	41
Cell recovery and drying	22
Product storage	10
(c) <i>ICI plc. Single-cell protein plant</i> (Smith, 1980)	
Raw materials	3
Storage and packing	12
Off-site services	16
On-site services	11
Fermentation	14
Compression	9
Dewatering	19
Drying	12
Effluent treatment	4

Class 3 microbiological cabinet for fermenters of up to 50 dm³. Buckland (1992) has stated that it now costs as much to build a 3 m³ facility for a recombinant protein which will need extensive containment provisions as for a 2000 m³ scale facility for an antibiotic. In a containment facility only 30% may be usable space. The remaining 70% of the building is needed to accommodate and service the facility.

Reisman (1993) has stressed the need for the correct level of containment. More stringent requirements may exist in a laboratory than are strictly needed. The same high level of containment in a production facility may be onerous, costly and unnecessary. Containment needs should be determined jointly by a group of appropriately trained staff.

The designing of vessels which can be converted for multi-use (Reisman, 1993) may have economic advantages, but descriptions of such vessels except at pilot scale are not common (Hambleton *et al.*, 1991).

The life of a proposed plant has to be predicted. MacLennan (1976) has estimated the life of a single-cell protein plant to be 10 years. This is probably the lower limit. Hamer (1979) claimed that a planned SCP plant life of 15 years is typical. Acetone and butanol have been produced in 25-year-old vessels (Spivey, 1978), while in long-established breweries in Great Britain, equipment of 50 to 100 years old is still used.

Allowance must be made for the service provisions for a fermentation process. In a moderate sized antibiotic plant, there may be the need for the provision of 5 tonnes h⁻¹ of steam, 5000 kW h⁻¹ of electricity, 57,000 m³ h⁻¹ of compressed air and 200,000 dm³ h⁻¹ of water (Hastings and Jackson, 1965). When producing 1 tonne of acetic acid, 480 m³ of cooling water, 10 m³ of process water, 12 tonnes of steam and 570 kW of electricity would be utilized (Pape, 1977).

MEDIA

The cost of the various components of a production medium can have a profound effect on the overall cost of a fermentation process, since these account for 38 to 73% of the total production cost (Table 12.1). The organic-carbon source in microbial processes is usually the most expensive component contributing to the cost of the process. Ratledge (1977) has made a detailed analysis of annual price and availability of major carbon substrates. The price of a natural material may fluctuate due to other competing demands and the annual variation in the quantity harvested. Big capital investment may be tied up in natural materials if they

are seasonal and require storage. Hastings and Jackson (1965) stated that up to 23×10^6 dm³ of molasses may have to be stored for an industrial alcohol-production process. A particular material may be selected because it is cheap locally, rather than the best substrate (Calam, 1967).

In a cost study analysis for tissue plasminogen activator by a mammalian-cell process, fermentation materials accounted for 75% of the total raw materials cost (Datar *et al.*, 1993). Of this cost, calf serum was estimated to contribute 73% or 55% of the total cost. However, it has been possible to develop a serum-free medium for economic production of antibodies from hybridoma cells (Maiorella, 1992).

In the search for suitable nutrient components for media, six criteria were considered in Chapter 4 which ought to be satisfied whenever possible. Ratledge (1977) has stressed the need to note the amount of carbon in a carbon substrate when costing. Higher yields might be achieved by changing from a carbohydrate to an alternative carbon source, but increased aeration and/or agitation rates may be necessary with the change in substrate. The cost of this extra provision must therefore be less than the savings from the change of substrate if the process is to be feasible. When cheapness is added as a further restraint the number of potential major nutrients which could be used on an industrial scale is limited. Carbohydrates from beet, sugar cane or grain are the major carbon and energy sources in most media and comply with the requirements of economy and those stated in Chapter 4. In 1993, sugar and starch based fermentation substrates were available within the EEC for approximately £200 tonne⁻¹. However, these are commodities whose cost fluctuates according to supply and demand. It is therefore worthwhile to develop a series of cost optimized media formulations so that the most cost effective growth medium can always be used when necessary (Winkler, 1991).

During the 1970s there was considerable interest in using petrochemicals as substrates for SCP production as protein animal feed by a number of major chemical and petroleum companies (Sharp, 1989). A number of processes were developed using methane or methanol as the main carbon substrate. None are now being operated. Major factors which contributed to making these processes uncompetitive included the increased cost of the substrate (see Chapter 4) and the availability of cheaper alternative animal feeds.

A variety of waste materials would seem to be potential cheap carbon sources. Unfortunately, it has been

shown that their use is very restricted because they cannot compete economically with conventional substrates. This may be due to a number of possible reasons including variability of the material, impurities which make downstream processing more difficult, high water content making transport costly, geographical location, quantities produced and limited seasonal availability. The economics of biomass production from whey have been reviewed by Meyrath and Bayer (1979). The possible use of waste substrates may depend on the cost of alternative methods of disposal or on the availability of government grants to diminish pollution (Perlman, 1970).

Mineral components normally constitute a smaller part of the cost of media, e.g. they account for 4 to 14% of the manufacturing cost of single-cell protein (Cooney and Makiguchi, 1977). Although feed grades of phosphates are more expensive than fertilizer grades, they do not contain impurities such as iron, arsenic and fluoride. This is an important consideration in the production of foods and drugs (Litchfield, 1977). The hydroxides and sulphates of potassium, magnesium, manganese, zinc and iron are preferred to the chlorides to minimize corrosion of stainless steel. The source of basic materials can cause considerable variation in product yield. Corbett (1980) compared six samples of calcium carbonate, and found that five of them reduced the titre of penicillin G in a production medium.

Problems concerned with the storage, handling and mixing of media should not be neglected. Powders must be kept in dry conditions because of the possibility of substances becoming rock-like or glutinous. Some bulk liquids with a high solids content need to be kept warm to prevent them solidifying, e.g. glucose and corn-steep liquor. If storage temperatures are too high there could be degradation. It is also vital for workers to follow instructions for media preparation very carefully to prevent 'lumpy' media, etc. (Corbett, 1980).

AIR STERILIZATION

The problems associated with producing large volumes of sterile air for aerobic fermentations are unique (see Chapter 5). Although sterilization by heating is technically possible, it has generally been regarded as too costly for full-scale operation (Cherry *et al.*, 1963), although it might be used in the treatment of exhaust gases (Walker *et al.*, 1987).

Absolute fixed-pore membrane systems using pleated membranes of PTFE are now widely used in the fermentation industry (Chapter 5) and have proved to be

very reliable. This is very important when considering the costs associated with loss of fermentation batches due to contamination and production downtime due to filter failure. Banks (1979) has reported that a contamination probability of 1 in 1000 is economically acceptable for microbial batch fermentations, while in large scale animal-cell culture processes contamination rates as low as 2% are now achievable (Spier, 1988).

Operating costs will be based on the estimated life of the filters. Factors to consider include the cost of replacement filters or filter materials, servicing and labour. Even if the filters could be cleaned there must be an allowance for depreciation due to normal wear and tear. Savings may also be made by introducing series filtration whereby the major part of the foreign matter from the air stream is taken out by varying degrees of coarse filtration, thus reducing renewals of the more expensive high-efficiency filter media such as membrane filters.

The treatment of fermenter exhaust gases to satisfy containment requirements is also important (see also Chapters 5 and 7). Treatment is normally by filtration with 0.2- μm hydrofilters, but in-line incinerators may be an alternative approach. Filtration is usually cheaper but it may be necessary to supplement filtration with incineration depending on the process and scale (Walker *et al.*, 1987).

HEATING AND COOLING

Ideally there should be no heating or cooling at any stage in a fermentation process, but because this is virtually impossible, heat should be conserved and cooling minimized by careful process design. A fermentation may include the following heating or cooling stages:

1. Sterilization or boiling of the medium to 100° or above followed by cooling to 35° or below.
2. Heating the fermenter and ancillary equipment to sterilize it, followed by cooling.
3. Heat may be generated during the fermentation. This heat output has to be removed by cooling to maintain the growth temperature of the micro-organism within prescribed limits.
4. After harvesting, heat may be required to remove water from the product.

Cooling requirements will be influenced by the size and type of an individual process (Chapter 7). British

Petroleum Ltd. estimated that the cooling requirements for a 100,000 tonnes annum⁻¹ single-cell protein plant to be 110×10^6 kcal h⁻¹ using n-alkanes or gas-oil as the primary substrate (Litchfield, 1977). To reduce cooling requirements, the specific energy input may be minimized through the use of air-lift fermenters (Schugerl *et al.*, 1978). Cooling equipment has been estimated at 10 to 15% of the investment cost for single-cell protein (Moo-Young, 1977; Schugerl *et al.*, 1978). Another way to minimize cooling costs is to use micro-organisms with higher optimum growth temperatures, if it is feasible. The selection and use of thermophiles and thermotolerant organisms would have obvious advantages to reduce cooling demands (Chapter 3).

AERATION AND AGITATION

Nearly all fermentations require some form of mixing to maintain a constant environment, and many also need aerating (see also Chapter 9). Fermentations may be broadly classified into:

1. Fermentations which are anaerobic where oxygen is undesirable, e.g. acetone-butanol.
2. Fermentations which have a minimal oxygen demand, e.g. ethanol.
3. Fermentations which have a high oxygen demand, e.g. antibiotics, acetic acid, single-cell protein.

In categories 1 and 2, aeration is not generally regarded as a major economic consideration. During an acetone-butanol fermentation carbon dioxide and hydrogen are evolved. Once this gas production starts it will help to maintain anaerobic conditions and stir the mass of broth without the need for mechanical agitation. Anaerobic conditions are achieved initially in a

production fermenter by maintaining a positive pressure of filtered carbon dioxide and hydrogen obtained from another established fermentation (Beesch, 1952).

For ethanol production, the yeast inoculum in the vessel is initially dispersed in the medium by compressed air or by mechanical stirring. Aeration or agitation is stopped once the biomass concentration reaches a predetermined level. A vigorous anaerobic fermentation commences, and the evolution of carbon dioxide bubbles stirs the contents of the vessel and disperses the cells in the medium so that mechanical agitation is unnecessary. In this process aeration and agitation are considered to be a minor component of the total production costs.

Fermentations having a high oxygen demand must be agitated with sufficient power to maintain a uniform environment and to disperse the stream of air introduced by aeration. In an early reference it was stated that the cost of energy necessary to compress air for yeast production proved that a considerable amount (10 to 20%) of the total production expenses was due to aeration (de Becze and Liebmann, 1944). Swartz (1979) has reported that the mixing costs in a penicillin fermentation are > 15% of the total production costs. Energy consumption for a stirred aerobic fermentation to provide agitation, air compression and chilled water is approximately 8.2 kW m⁻³ (Curran and Smith, 1989). Assuming an electricity cost of \$0.07 kW⁻¹, a 6-day antibiotic fermentation in a 100-m³ fermenter with a 1-day turnaround would use \$8,000 of power (Royce, 1993).

In single-cell protein processes, the carbon substrate yield coefficient is the most critical physiological factor (Hamer, 1979). It is also well documented that much higher carbon-substrate yield coefficients are obtained with methane or n-alkanes instead of carbohydrates (see Table 12.6). Unfortunately, cells grown on hydrocarbons have greater oxygen requirements. The oxygen requirements of a hydrocarbon yeast fermentation is

TABLE 12.6. *Effect of substrate and yield coefficients on SCP operating costs (Abbott and Clamen, 1973)*

Substrate	Substrate costs		O ₂ transfer costs (£ lb ⁻¹ cells)	Heat removal costs (£ lb ⁻¹ cells)	Combined costs (£ lb ⁻¹ cells)
	(£ lb ⁻¹ substrate)	(£ lb ⁻¹ cells)			
Maleate (waste)	0	0	0.46	0.75	1.2
Glucose (molasses)	2.0	3.9	0.23	0.54	4.7
n-Paraffins	4.0	4.0	0.97	1.4	6.4
Methanol	2.0	5.0	1.2	1.9	8.1
Ethanol	6.0	8.8	0.75	1.3	10.9
Acetate	6.0	16.7	0.62	1.1	18.4

almost triple that of a yeast fermentation grown on a carbohydrate substrate and producing an equal quantity of cells (Darlington, 1964; Chapter 4). Therefore, if there is to be effective utilization of a hydrocarbon substrate, which can account for over 50% of total production costs (Litchfield, 1977), the production fermenter must have a high oxygen-transfer capacity. The demands on fermenter design are further complicated by the hydrocarbon fermentation being highly exothermic, which necessitates the provision of good cooling facilities if a constant temperature is to be maintained in the fermenter.

A few companies developed SCP processes using mechanically stirred fermenters with sparged air. BP Ltd. constructed vessels of up to 1000 m³ capacity for their n-alkane process in their Sardinian Ital protein project (Levi *et al.*, 1979). In the Swedish Norprotein process it has been estimated that the total utilities costs, which included aeration and agitation (1978 prices) for 100,000 tons year⁻¹ of SCP would only be 16% of total production costs (Mogren, 1979).

A number of companies, including ICI plc (Taylor and Senior, 1978) and Hoechst (Knecht *et al.*, 1977), decided to develop fermenters based on the air-lift principle (Chapter 7; Hamer, 1979; Levi *et al.*, 1979; Sharp, 1989). The main advantages of these fermenters are simpler design and reduced energy and cooling water costs. Since the energy supplied to an air-lift fermenter is only supplied with the air, it is crucial to obtain a fermenter design which minimizes the energy requirement for biomass production yet creates high oxygen-transfer facilities to ensure efficient substrate utilization. In the ICI plc process, the estimated manufacturing costs for all utilities were 14%, with aeration accounting for 70% of fermentation utilities costs (Moo-Young, 1977).

BATCH-PROCESS CYCLE TIMES

In a batch process productivity must be determined for the complete process cycle. Here productivity is defined as grams of product dm⁻³ h⁻¹. This productivity is based on a combination of the time for the actual fermentation and the time to prepare the fermenter ready for the next run. Heijnen *et al.* (1992) estimated this to take 15–25 h. Thus the total time for a fermentation may be calculated (Wang *et al.*, 1979) as:

$$t = \frac{1}{\mu_m} \cdot \ln \frac{X_f}{X_0} + t_T + t_L + t_D$$

where μ_m = maximum specific growth rate,
 X_0 = initial cell concentration,

X_f = final cell concentration,
 t_T = turn-around time (washing, sterilizing, filling with media),
 t_D = delay time until inoculation,
 t_L = lag time after inoculation.

The overall productivity P is given by:

$$P = \frac{X_f}{\frac{1}{\mu_m} \ln \frac{X_f}{X_0} + t_T + t_D + t_L}$$

It will be possible from this equation to determine the effects of process changes on the overall productivity. A larger initial inoculum would increase X_0 and shorten the process time. Actively growing inocula would reduce the lag time t_L . Aspects of this problem have been discussed in Chapter 6. It is also worthwhile to isolate faster-growing organisms and/or higher yielding strains (Chapter 3). In many processes the growth phase and/or the production phase has been extended by the use of fed-batch or continuous feed (Chapter 2) with improved productivity.

Richards (1968a,b), Geysen and Gray (1972) and Stowell and Bateson (1984) have given details for determining maximum production at minimum cost and the optimum time for harvesting (Fig. 12.1).

In fermentations with short growth cycles such as bakers' yeast (14 to 24 hours), the turnaround time will be as important as the time between inoculation and

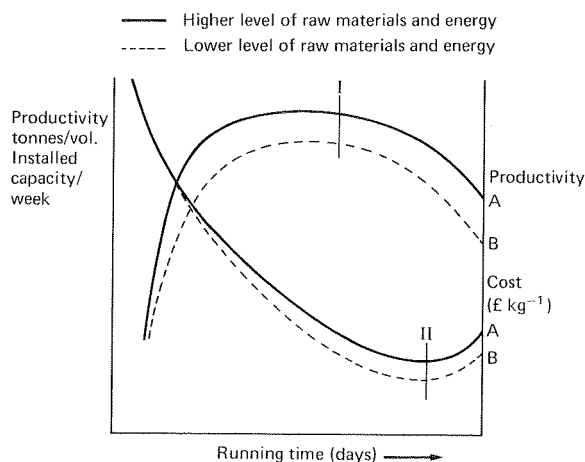


FIG. 12.1. Fed-batch process—effect of running time on productivity and product cost (Stowell and Bateson, 1984). Fermentation should be terminated at I for maximum productivity, II for minimum cost. Productivity is defined as the units of product generated per unit of fermenter volume in a given time.

harvesting. When the production cycle is long, as with penicillin (6 to 7 days), a few extra hours for turnaround will have little influence on overall productivity.

CONTINUOUS CULTURE

At the present time, very few large-scale continuous-culture processes are being operated. These are primarily for the production of microbial biomass, glucose isomerase, buttermilk souring and yoghurt (Heijnen *et al.*, 1992).

It is appropriate at this stage to compare batch-culture productivity and continuous culture productivity. Wang *et al.* (1979) derived an equation to quantify this relationship:

$$\frac{\text{Continuous-culture productivity}}{\text{Batch-culture productivity}} = \frac{\ln(X_m/X_0) + \mu_m t_L}{(X_m - X_0)/X_m} D_c Y$$

where μ_m = maximum specific-growth rate,
 X_0 = initial cell density,
 X_m = maximum cell density,
 t_L = turn-around time,
 D_c = critical dilution rate,
 Y = cellular yield coefficient for the limiting nutrient.

In an example they used an inoculum size of 5% ($X_0/X_m = 0.05$), a process turn-around time of 10 hours, a cellular yield of 0.5 g cells per g of substrate and a final cell concentration of 30 g dm⁻³. Productivity was then calculated for a series of maximum specific growth rates (Table 12.7). It is clear from these data that the faster the growth of the organism, the more favourable is a continuous process over a batch process.

When assessing feasibility of a continuous process for product formation it is necessary to know: volumetric productivity, conversion yield of product from the

most expensive substrate in the medium and the product concentration (Wang *et al.*, 1979). In processes for cell biomass, some alcohols and organic acids where the major production cost is at the fermentation stage, volumetric productivity and conversion yield will be most important. A continuous process may be more economic than a batch process if higher productivities at higher efficiencies can be achieved. Unfortunately, in some processes, the final product concentration in the effluent broth from a continuous culture will be less than that obtained in a batch process, which will create a need for greater concentration at the recovery stage.

Heijnen *et al.* (1992) argue that low dilution rates are favoured if high product concentrations are to be obtained as recovery cost may be 50–80% of total costs. However, the concentration preference is not valid in processes which can start with a simple concentration step (SCP or intracellular glucose isomerase) or with products that need no concentrating (beer, yoghurt and buttermilk).

In a production plant, continuous culture offers the advantages of constant flow, product quality and simple automation and control. Disadvantages are due to specific production facilities that cannot be used for other purposes, lack of continuous recovery techniques and lack of constant market demand (Heijnen *et al.*, 1992).

The continued increase in efficiency of fed-batch culture processes for antibiotics and other non-growth-associated products makes manufacturers reluctant to make radical alterations to established processes such as the introduction of continuous culture (see also Chapter 2).

RECOVERY COSTS

The costs of product recovery and purification are rarely quoted, though in some processes they are obviously considerable. It is now accepted that cost analy-

TABLE 12.7. Comparison of productivity in batch and continuous culture (Wang *et al.*, 1979)

Maximum specific growth rate in batch culture (h ⁻¹)	Continuous productivity
	Batch productivity
0.05	0.09
0.10	0.21
0.20	0.53
0.40	1.5
0.80	4.6
1.0	6.8
1.2	9.5

ses of 'older' fermentation processes split approximately 1:1 between fermentation costs and isolation/purification costs (Reisman, 1993). However, the change in the last 10 years to use recombinant organisms to produce exotic compounds at extremely low titres has meant that the fermentation/isolation-purification split may be 1:8 or 1:10. This means that the fermentation may only be 10% of the costs, while the recovery accounts for 90%. In this case the correct choice of the recovery-purification procedure can be crucial to the success of a process and early evaluation of alternative techniques may be very important.

Stowell and Bateson (1984) identified a number of factors contributing to these costs:

1. Yield losses, even if only modest, are certain to occur at each stage of the recovery process.
2. High energy and maintenance costs associated with running filtration and centrifugation equipment.
3. High costs of solvents and other raw materials used in recovery and refining of products.

Atkinson and Mavituna (1991) reported losses of 8% for citric acid and 4% for penicillin G in the recovery and purification stages before conversion to the potassium salt in production processes. They also stressed the importance of trying to reduce the number of downstream stages as much as possible to reduce capital and operating costs.

It is thought that depreciation, return on capital and maintenance can account for over 80% of the overall cost for a large-scale rotary filtration or centrifugation plant (Atkinson and Mavituna, 1983). However, it is considered that removing cells by filtration is less energy consuming than by centrifuging. If filter aids are to be used, in the most economical way, this will still add £9 tonne⁻¹ for a product at a 10% concentration in the broth.

One of the main factors affecting centrifuge economics is the size of the particle to be separated (Asenjo, 1990). Filtration costs are less dependent on particle size. At a particle size greater than 1 to 2 µm, centrifugation is more economical. Below this size ultrafiltration becomes more economical.

Taksen (1986) has given a case study for a moderate value product (£1.50 to 3.00 per kg) requiring the processing of 200,000 dm³ of broth per day by rotary vacuum filtration or centrifugation. Capital investment for the rotary filter would be £500,000 with an operating cost of 1.15 p per dm³ of broth. A three-stage

centrifuge would cost £667,000 and broth could be processed at 0.62p per dm³. Because rotary filters were already available, it was decided that the new capital investment could not be justified, although the broth would cost more to process. In another case study on ultrafiltration vs. evaporation to pre-concentrate a moderately high-molecular-weight product, the cost of water removal by membrane techniques was estimated to be 40% lower than by evaporation.

Maiorella *et al.* (1984) have compared extraction processes in 11 alternative ethanol fermentation processes. Selective ethanol removal by flash distillation was thought to be the most economic technique.

When a product may be made by a microbial process or obtained from an alternative source there are cost limits on product recovery to be considered. Atkinson and Mavituna (1991) estimated:

1. A limit of about £40 tonne⁻¹ for ethanol selling at £220 tonne⁻¹ produced at 7% w/v in broth from a local source of carbohydrate.
2. A limit of about £100 tonne⁻¹ for SCP selling at £300 tonne⁻¹ produced from a petroleum-based substrate to give a 3% w/v yield.
3. A limit of about £300 tonne⁻¹ for organic acids or glycol selling at £700 tonne⁻¹, produced at 10% concentration in a carbohydrate medium.

When high-value end-products have been produced it has been acceptable to use relatively large weights of filter aid to achieve initial clarification to remove small amounts of solids. The solvents used in subsequent extraction have then been recovered in a high energy-consuming distillation plant (Atkinson and Sainter, 1982). In this case, the manufacturer has only to make his product as economically as those of other fermentation companies.

Before certain products may be marketed the extraction/purification procedure will have to be validated (approved) by the FDA (U.S.A.) or similar regulatory bodies. Any changes in extraction procedures will have to be checked for revalidation, which will incur further costs (Reisman, 1993). Therefore, if validation is to become an issue in processing it may be worthwhile having alternative procedures validated at an early stage in development.

WATER USAGE AND RECYCLING

Many fermentations have a high daily water usage (Table 12.8). As charges for water increase, many of

TABLE 12.8. Daily water usage in fermentation processes

Industry	m ³ of water used day ⁻¹	Reference
Maltings	230	Askew (1975)
Brewing	10,000	
Distilling	320	
Antibiotics	245	Hastings and Jackson (1965)
Antibiotics	5,200	
Acetic acid	700	
Single cell protein (methanol substrate)	4,000 to 12,000	Pape (1977)
Yeast (alkane substrate)	18,200	Taylor and Senior (1978)
Bacteria (methanol substrate)	45,500	
Bacteria (methane substrate)	18,200	

these processes will become vulnerable to cost escalation because of the relatively large volumes of water required per unit volume of product.

There is now a widespread interest in reducing overall consumption. In a bacitracin plant described by Inskeep *et al.* (1951), the water from the mash cooler was collected and reused to charge the mashing vessels and wash the fermenters. Water from the cooler coils was used to wash down the discharge cake from the filter presses. Bernstein *et al.* (1977) designed a 'closed-loop' system for fermenting cheese whey in which effluents were completely recycled.

Recycling of water was an integral stage of large scale SCP processes developed during the 1970s to minimize water consumption, reduce effluent treatment costs and reduce media costs by recycling of spent media (Sharp, 1989). When ICI plc's SCP Pruteen plant was operating, it was designed to recycle most of the fermenter medium water (Ashley and Rodgers, 1986). Under optimum conditions they claimed that the water loss could be reduced to 3% of the flow through the fermenter using water recovery systems.

EFFLUENT TREATMENT

Before deciding on the most economic form of treatment for wastes it is important to make a factory survey. The information should include the water volume, the organic and solids loading, range of pH variation, nutrient level, temperature fluctuation, and the presence of any toxic compounds. It will also be necessary to consider company finance policies, the site location and government legislation for waste disposal (see also Chapter 11).

In the majority of fermentation processes it is impossible to dispose of effluents at zero cost. Whether the

waste is incinerated, dumped on waste land, or discharged to sewers, rivers or tidal waters, some expenditure will be necessary for treatment that ensures that minimal harm is done to the environment.

Since the 1980s the European Community has adopted a number of Directives to reinforce earlier legislation for the protection and improvement of inland and marine water quality. The standards of these Directives and consent for discharge are implemented in the United Kingdom through the monitoring of the National Rivers Authority and the water companies in England and Wales, the River Purification Boards in Scotland and the Department of the Environment in Northern Ireland (Brown, 1992). Costs to meet the requirements of these Directives will need to be included in process costings.

The various alternative disposal procedures may be compared using economic considerations. Pape (1977) claimed that the cheapest treatment method was controlled dumping, followed by waste incineration or dumping in salt mines. The most expensive method was biological degradation in a waste-water-treatment plant. The last method has often to be used because the effluents usually contain only a few percent of organic matter which would be costly to separate, concentrate and incinerate.

The possibility of direct disposal of pharmaceutical waste into the sea is now very restricted, especially if the waste is untreated, even though many of the large fermentation plants in the United Kingdom are in coastal locations. In 1972, Jackson and Lines stated that a pipeline of over 2.8 km overland and 2.8 km on the sea bed at a cost of £350,000 would be needed to dispose of 3000 dm³ day⁻¹ of untreated antibiotic waste. The other options are to discharge the effluent direct to the sewers and pay a charge, to treat the waste in the plant, or to operate a combination of the two. Sewage works' charges for treating effluents have

increased 1000% in less than 5 years in some instances (Forage, 1978). Those plants which treat all or part of their effluents have discovered that energy costs have risen and sludge disposal is more costly and difficult. Ripley (1979) estimated the costs for a treatment plant for a whisky distillery producing $4,500,000 \text{ dm}^3 \text{ year}^{-1}$ of proof whisky to have a capital cost of £75,000 and operating costs of £9000 year^{-1} . Power was calculated at 0.9 kW kg^{-1} of BOD removed while dosage of nutrients was $£0.03 \text{ kg}^{-1}$ of effluent BOD.

Avermectin is an antihelmintic compound produced by *Streptomyces avermitilis* (Omstead *et al.*, 1989). During development of this compound it was recognized that it was very potent and could have a potential impact on aquatic fauna. All possible exit streams from the process at the factory, both fermentation and downstream purification, are therefore captured and chemically degraded. In this case, environmentally safe waste treatment is a major component in production costs. An alternative method for disposing of wastes of this type would be to absorb all waste streams in a suitable material and incinerate this and solid wastes at an appropriate temperature (see also Chapter 11).

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Index

- Abrasives, agitation 293
- Acetators, fermenters 199
- Acetone 7
- Activated sludge processes 323
- Aeration 341
 - effect of microbial products 266
 - fermenters 178, 183
 - scale up in stirred tank reactors 270
- Aerobic fermenter, vinegar generator 7
- Aggregation 287
- Agitation 341
 - fermenters 178
 - scale up in stirred tank reactors 270
 - with abrasives, cell disruption 293
- Agitator
 - impeller in a fermenter 178, 261
 - shaft power 220
- Air
 - filter sterilization 139
 - lift
 - fermenter 202, 208
 - reactors, scale up 272
 - sterilization 340
- Alkali treatment, cell disruption 295
- Amino acids 7
 - wastes 327
- Anaerobic
 - digesters 325
 - digestion 324
 - filters 326
 - upflow sludge blankets 326
 - treatment, fermentation wastes 324
- Animal cell
 - culture, fermenters 168, 205
 - media 115
- Animal feed, production 2
- Antibiotic yields, improvements 58
- Antibiotics 5
- Antifoams 108
 - effect on oxygen transfer 267
- Arginine biosynthesis, control of in *C. glutamicum* 51
- Aseptic
 - conditions
 - achievement 185
 - maintenance 185
 - fermentation 7
 - inoculation
 - fermenters 162, 187
 - operation, fermenters 168
- Automatic control 228
- Auxotrophic mutants, isolation 61
- Auxotrophs 50
 - production of primary metabolites 50
- Bacillus subtilis* 52
 - control of purine nucleotide synthesis 52
- Bacterial processes, inoculum development 153
- Baffles, fermenters 183
- Baker's yeast 7
 - production 1
- Barley malt, carbohydrate composition 99
- Basket centrifuge 288
- Batch
 - culture, kinetics 13
 - filters 283
 - process cycle times 342
 - processes 13
 - sterilization 128
 - design 129
 - methods 130
 - scale up 130
- Benzyl penicillin, biosynthesis 62
- Bingham plastic rheology 253
- Biological treatment, fermentation wastes 320
- Biologically aerated filters 321
- Biomass
 - concentration, control 268
 - feedback 19
 - external 19
 - internal 19
 - microbial 1
 - production, continuous culture 25
 - productivity, kinetics 21
- Biosynthetic pathways, control of 45
- Bovine somatostatin 5
- Breweries
 - large-scale 6
 - waste by-products 327
- Brewing, continuous 24
 - cylindro-conical vessels 25, 152, 201, 337
- Buffers, in media 105
- Bugmeter 151, 222
- Butanol 7
- By-products, fermentation wastes 326

Index

- Calf chymosin 5
- Carbohydrates
 - as carbon source 98
 - barley malt composition 99
- Carbon dioxide electrodes 226
- Carbon sources
 - carbohydrates 98
 - fats 99
 - hydrocarbons 99
 - in media 93, 97
 - oils 99
- Casson body rheology 254
- Cavitators fermenters 199
- Cell disruption 292
- Centrifugation 287
- Centrifuges
 - basket 288
 - decanter 289
 - disc-bowl 290
 - multichamber 289
 - perforated bowl basket 288
 - solid bowl scroll 289
 - tubular bowl 288
 - types 288
- Cepharmycin C 279
- Chelators, in media 104
- Chemical treatment, fermentation wastes 320
- Chiral catalysts 5
- Chromatography 301
 - adsorption 302
 - affinity 303
 - continuous 304
 - gel permeation 303
 - high performance liquid (HPLC) 304
 - ion exchange 302
 - reverse phase 303
- Chronological development, fermentation industry 8
- Citric acid 7
 - recovery 279
- Comparisons, batch with continuous culture in industry 21
- Components of fermentation processes 9
- Computer linked control systems 234
- Computers
 - in fermentation technology 234
 - process control 237
- Concerted feedback control 46
- Containment, fermenters 168
- Contamination, of fermentation process 123
- Continuous
 - brewing 24
 - culture 8, 342
 - biomass production 25
 - kinetics 16
 - enrichment techniques 38
 - filters 284
 - processes 13
 - sterilization 128
 - design 132
- Control
 - automatic 228
 - combination methods 231
 - complex systems 232
 - computer linked 234
 - integral 231
 - manual 228
 - of biomass concentration 268
 - of dissolved oxygen 222
 - of flow 217
 - of foam 221
 - of pH 225
 - of pressure 220
 - of specific oxygen uptake rate 269
 - of temperature 217
 - proportional 229
 - systems 214 228
- Controller gain 230, 231
- Controllers 232
 - two position 229
- Control loop 228, 229, 232, 233, 235, 269
- Cooperative feedback control 46
- Corn steep liquor, composition 100
- Corynebacterium glutamicum* 47, 50
 - control of arginine biosynthesis 51
- Cost of recombinant protein production 9
- Criteria
 - for inoculum transfer 149
 - for suitable inocula 147
- Cross flow filtration 285
- Crystallization 307
- Culture
 - collections, major 36
 - production 9
 - rheology, and $K_L a$ values 260
- Cumulative feedback control 46
- Cyclic fed-batch culture 29
- Cyclone column fermenter 205
- Cyclosporin 9
- Cyindro-conical vessel fermenter 201
- Data
 - analysis 236
 - logging 236
- Decanter centrifuge 289
- Deep jet fermenter 204
- Del factor 129
- Depth filters
 - design 144
 - theory 140
- Design
 - depth filters 144
 - filters 142
 - sterilization cycles 130
- Detergents, cell disruption 294
- Dilatant rheology 254
- Disposal, effluents 10, 317
- Distilleries, waste byproducts 326
- Drying, of product 305
- Dyno-Muhle KD5 disintegrator 293, 295
- Economics
 - effluent treatment 345
 - media 339
 - of fermentation equipment 336
 - of fermentation plants 336
 - recycling 344
 - water usage 344

- Effluents
 - disposal 10, 316
 - incineration 318
 - lagoons 317
 - landfill 318
 - seas and rivers 317
 - spray irrigation 317
 - to sewers 318
 - wells 318
 - from fermentation 315
 - treatment 316
- Electrodes
 - carbon dioxide 226
 - enzyme 227
 - microbial cell 227
- Energy source, media 97
- Enrichment liquid culture 37
- Enzyme
 - electrodes 227
 - inducers, used in fermentation processes 107
 - treatment, cell disruption 295
- Enzymes 7
 - commercial applications 3
 - microbial 2
- Epidermal growth factor 5
- Erythropoietin 8
- Escherichia coli* 5
- External feedback, kinetic description 20
- Extraction
 - liquid-liquid 296
 - of product 10
 - supercritical fluid 301
 - two-phase aqueous 300
- Factor
 - IX 5
 - VIII 5
- Fast metabolism 108
- Fats, as carbon source 99
- Fed-batch
 - culture
 - application 29
 - cyclic 29
 - fixed volume 28
 - uses 29
 - variable volume 27
 - processes 13
- Feed ports, fermenters 189
- Feedback control
 - concerted 46
 - cooperative 46
 - cumulative 46
 - multivalent 46
- Feeds, sterilization 137
- Fermentation
 - contamination 123
 - effluents 315
 - industry
 - chronological development 6
 - development of 5
 - history 5
 - media 94
 - oxygen requirements 243
 - process, components 9
 - processes
 - batch 13
 - continuous 13
 - fed-batch 13
 - range of 1
 - vessels, types 199
 - wastes, by-products 326
- Fermentations
 - commercially important 1
 - industrial media 93
- Fermenter
 - air supply, sterilization 186
 - exhaust air, sterilization 140
 - sterilization 137
 - stirrer glands 181
- Fermenters
 - acetators 199
 - aeration 178
 - agitation 178
 - air lift 202, 208
 - animal cell culture 168, 205
 - aseptic
 - inoculation 187
 - operation 168
 - baffles 183
 - basic functions 168
 - body construction 171
 - cavitators 199
 - construction materials 171
 - containment 168
 - cyclone column 205
 - cylindro-conical vessels 201
 - deep jet 204
 - exhaust gas sterilization 187
 - feed ports 189
 - foam control 189
 - hollow fibre chambers 209
 - magnetic drives 182
 - mechanical seal 181
 - microbial cell culture 168
 - microcarriers 208
 - packed glass bead reactors 209
 - packed tower 205
 - perfusion cultures 209
 - rotating disc 205
 - sampling 187
 - sensor probes 189
 - sparger 183
 - steam traps 192, 196
 - sterilization 186
 - stirred 205
 - stuffing box 181
 - temperature control 171
 - testing 198
 - tower type 200
 - valves 192
 - Waldhof type 199
- Filamentous fungi 5, 14, 160, 206
- Filter
 - aids 282
 - design 142

Index

- sterilization 137
 - of air 139
 - of media 139
- Filters
 - anaerobic 326
 - biologically aerated 321
 - continuous 284
 - cross-flow 285
 - plate and frame 283
 - pressure leaf 283
 - rotary vacuum 284
 - trickling 320
- Filtration 281
 - cross flow 285
 - tangential 285
 - theory 281
 - ultrafiltration 277, 278, 286, 304, 305, 307, 344
- Fixed volume fed-batch culture 28
- Flocculation 287
- Flow
 - control 217
 - measurement 217
- Fluid rheology 252
- Fluidized bed systems 322
- Foam
 - control 221
 - fermenters 189
 - sensing probe 221
 - separation 280
- Foaming 80, 81, 190, 221
 - patterns in fermentations 80, 81, 190, 221
- Foams
 - effect on oxygen transfer 267
- Fuel alcohol wastes 327
- Fungal biomass, human food 3
- Gas analysis
 - exit 224
 - inlet 224
- Gases, flow control 217
- Gassing out
 - dynamic method 249
 - static method 248
 - technique 248
- Gateway sensors 221, 236
- Genetic engineering processes 8
- Genetically engineered organisms 5
- Gibberellin 7
- Glutamic acid fermentation 47
- Glycerol 7
- Granulocyte colony stimulating factor 8
- Growth factors, in media 104
- Hazard categories 170
- Heat exchangers 6, 133, 233
- Heterologous protein production 72
- Hollow fibre chambers, fermenters 209
- Human
 - food production 2
 - growth hormone 8
 - serum albumin 5
- Hydrocarbons, as carbon source 99
- Idiophase 4
- Imipenem 9
- Impeller, agitator in a fermenter 178
- Induced mutants 45
- Inducers, in media 106
- Industrial
 - fermentation, oxygen requirements 243
 - fermentations, media 93
 - micro-organisms, improvement 66
- Industrially important micro-organisms, isolation 35
- Infection, resistant strains 81
- Inhibitors
 - in media 105
 - used in fermentation processes 107
- Inoculation
 - aseptic, of plant fermenters 162
 - from laboratory fermenter 162
 - from plant fermenter 163
 - from spore suspension vessel 162
- Inoculum
 - development
 - bacterial processes 153
 - baker's yeast 153
 - brewing 151
 - criteria 147
 - for yeast processes 151
 - media 148
 - mycelial processes 155
 - vegetative fungi 160
 - production media 148
 - spore 158
 - transfer, criteria 149
- Insulin 5, 8
- Integral control 231
- Interferon 5
- Internal feedback, kinetic description 19
- Ion
 - exchange 302
 - specific sensors 227
- Isoenzyme control 46
- Isolation
 - auxotrophic mutants 61
 - industrially important micro-organisms 35
 - medium 35
 - methods 37
 - resistant mutants 63
 - revertant mutants 65
- Ivermectin 9
- Kinetics
 - batch culture 13
 - continuous culture 14
- $K_L a$ values
 - and air flow rate 254
 - and culture rheology 260
 - and microbial biomass 261
 - and power consumption 257
 - determination 247
 - factors affecting 254

- Laboratory fermenter, inoculation 162
- Lactic acid 7
- Lag phase 13
- Legislation, on antibiotic production, USA 336
- Liquid
 - membranes 305
 - shear, cell disruption 292
- Liquids, flow control 218
- Load cells 215, 218, 221
- Log phase 13
- Lovastatin 9
- Low oxygen tolerant strains 82
- Lysine biosynthesis 62

- Magnetic drives, fermenters 182
- Manual control 228
- Market potential, for microbial products 335
- Mass spectrometers 227
- Measurement
 - gas 224
 - of dissolved oxygen 222
 - of flow 217
 - of microbial biomass 221
 - of pH 225
 - of pressure 219
 - of temperature 216
 - weight of fermenter 221
- Mechanical seal, fermenters 181
- Media
 - animal cell 115
 - antifoams 94, 99, 108-110
 - buffers 105
 - carbon sources 93, 97
 - chelators 104
 - energy source 97
 - examples 95
 - filter sterilization 139
 - formulation 9, 94
 - growth factors 104
 - inducers 106
 - industrial fermentations 93
 - inhibitors 105
 - inoculum development 148
 - inoculum production 148
 - metabolic regulators 105
 - minerals 102
 - nitrogen sources 93, 101
 - non-nutritional supplements 116
 - optimization 110
 - osmolality 116
 - oxygen requirements 107
 - pH 116
 - precursors 105
 - protein free 115
 - serum 115
 - serum free 115
 - sporulation on 155
 - sterilization 123
 - trace elements 116
 - vitamin sources 104
 - water component 96
- Metabolic regulators, in media 105
- Metabolite productivity, kinetics 22
- Metabolites, microbial 3
- Micro-organism, optimum temperature 35
- Microbial
 - biomass 1
 - and $K_L a$ values 261
 - estimation of 221
 - cell culture, fermenters 168
 - cell electrodes 227
 - cultures, oxygen supply 246
 - enzymes 2
 - metabolites 3
 - products, effect on aeration 266
 - Microcarriers, fermenters 208
 - Microorganisms, potential for industry 332
 - Minerals, in media 102
 - Monoclonal antibodies 9
 - Morphologically favourable strains 82
 - Multichamber centrifuge 289
 - Multistage
 - chemostat 19
 - systems 19
 - Multivalent feedback control 46
 - Mutants
 - auxotrophic 61
 - induced 45
 - lack of feedback inhibitor production 48
 - lack of feedback inhibitor recognition 52
 - lack of repressor production 48
 - lack of repressor recognition 52
 - resistant 65
 - revertant 65
 - Mycelial processes, inoculum development 155
 - Myeloid colony stimulating factors 8

 - Native microbial product improvement 74
 - Near infra red spectroscopy 227
 - Neural network 239
 - New fermentation product-producing strains 83
 - Nitrogen sources
 - choices 102
 - in media 93, 101
 - Non-nutritional supplements, media 116
 - Non-foaming strains 81
 - Nozzle sparger, fermenters 184
 - Nutrient quality, sterilization 127

 - Oils, as carbon source 99
 - Optimization, of media 110
 - Optimum temperature 35
 - Orifice sparger, fermenters 184
 - Osmolality, media 116
 - Osmotic shock, cell disruption 295
 - Oxygen
 - availability in medium
 - antifoams 108
 - fast metabolism 108
 - rheology 108
 - balance technique 251
 - dissolved
 - as water quality indicator 314
 - control 222
 - measurement of 222

Index

- requirements
 - in media 107
 - of industrial fermentation 243
- supply and demand 267
- supply, to microbial cultures 246
- transfer
 - effect of antifoams 267
 - effect of foams 267
- uptake rate, control 269
- Packed
 - glass bead reactors, fermenters 209
 - tower fermenter 205
- Parasexual cycle 66
- Pellet formation 160, 161, 266
- Penicillin
 - G recovery 279
 - production 7
 - recovery 7
- Penicillium chrysogenum* 62
- Perfusion cultures, fermenters 209
- Permeability modification 47
- pH
 - control 225
 - measurement of 225
 - media 116
- Physical treatment, fermentation wastes 319
- Plant fermenter
 - inoculation 163
 - aseptic inoculation 162
- Plate and frame filters 283
- Porous sparger, fermenters 184
- Potable alcohol 6
- Power consumption
 - and $K_L a$ values 257
 - and operating variables 258
- Precipitation 280
- Precursors
 - in media 105
 - used in fermentation processes 106
- Pressure
 - control 220
 - cycle fermenters 8
 - jet fermenters 8
 - leaf filters 283
 - measurement 219
 - pressure gauges 220
 - safety valves 192, 196, 220
- Primary
 - catabolic routes 5
 - metabolites 4
 - improved levels 45
 - products of metabolism 4
- Process
 - control, computers 237
 - management 215
 - sensors, control functions 215
 - variables, measurement 215
- Product
 - extraction 10
 - recovery 35
 - yield 35
- Production of primary metabolites, auxotrophs 50
- Products of primary metabolism 4
- Proportional control 229
- Prostaglandins 5
- Protein
 - free media 115
 - production, heterologous 72
- Protoplast fusion techniques 68
- Prototrophs 50
- Pruteen animal feed, ICI 8
- Pseudoplastic rheology 253
- Purification
 - of cephamycin C 279
 - of citric acid 279
- Purine nucleotide synthesis, control of in *B. subtilis* 52
- Recombinant
 - DNA techniques 70
 - products 4
 - protein production, cost 9
 - proteins 8
 - systems 66
- Recovery
 - costs 343
 - crystallization 307
 - of citric acid 279
 - of penicillin G 279
 - of product from fermentation broth 278
 - solvent 299
- Redox potential 226
- Removal, of solid matter from fermentation broth 279
- Resistant mutants, isolation 63
- Reverse osmosis 305
- Revertant mutants, isolation 65
- Rheology 108
 - Bingham plastic 253
 - Casson body 254
 - dilatant 254
 - fluid 252
 - pseudoplastic 253
- Rotary vacuum filters 284
- Rotating
 - biological contactors 322
 - disc contactors 322
 - disc fermenters 205
 - drums 322
- Saccharomyces cerevisiae* 5
- Safety valves 220
- Sampling, fermenters 187
- Scale
 - down 269, 272
 - up 269
 - aeration 270
 - agitation 270
 - air lift reactors 272
- Secondary
 - compounds of metabolism 4
 - metabolites 4
 - improved yield 57
- Sensor probes, fermenters 189
- Sensors 215
 - in line 215

- ion specific 227
- off line 215
- on line 215
- Serum
 - free media 115
 - media 115
- Shear
 - liquid 292
 - solid 293
- Single cell protein, production 1
- Site surveys 314
- Solid
 - shear, cell disruption 293
- Solvent
 - fermentations 7
 - recovery 299
- Sparger
 - fermenters 183
 - nozzle 184
 - orifice 184
 - porous 184
- Spore
 - inoculum 158
 - suspension vessel, inoculation 162
- Sporulation, type of media 155
- Stable strains 80
- Starter cultures 7
- Steam traps, fermenters 192, 196
- Sterilization 9
 - air 340
 - batch 128
 - continuous 128
 - cycles, rapid method for design 130
 - feeds 137
 - fermenter air supply 186
 - fermenter exhaust air 140
 - fermenter exhaust gas 187
 - fermenters 137, 186
 - filter 137
 - nutrient quality 127
 - of media 123
- Steroid transformations 7
- Steroids 5
- Stirred
 - fermenters 205
 - tank reactors 270
 - glands, fermenter 181
- Stirring rate 220
- Stoichiometry 94, 96, 243
- Strain
 - development, new fermentation products 83
 - improvement 333
 - improvement programme 59
- Strains
 - infection resistant 81
 - low oxygen tolerant 82
 - media-component-resistant 81
 - morphologically favourable 82
 - non-foaming 81
 - stable 80
- Stuffing box, fermenters 181
- Sulphite oxidation technique 248
- Tangential filtration 285
- Temperature
 - control 217
 - fermenters 174
 - measurement 216
- Thermistors 216
- Thermometers
 - electrical resistance 216
 - mercury in glass 216
- Tower fermenter 200
 - brewing 25
 - effluent treatment 321
- Trace elements, media 116
- Transformation processes 5
- Treatment processes, fermentation wastes 318
- Trickling filters 320
- Trophophase 4
- Tubular bowl centrifuge 288
- Two position controllers 229
- Ultrafiltration 304
- Ultrasonication, cell disruption 292
- Upflow anaerobic sludge blankets 326
- Vaccine fermentations 8
- Valves
 - ball 193
 - butterfly 194
 - check 195
 - diaphragm 194
 - gate 192
 - globe 192
 - needle 193
 - pinch 194
 - piston 192
 - plug 193
 - pressure control 196
 - pressure reduction 196
 - pressure retaining 196
 - safety 196, 220
- Variable volume fed-batch culture 27
- Vegetative fungi, inoculum development 160
- Vinegar 6, 7
 - generator, as aerobic fermenter 7
 - production 5
- Vitamins 7
- Waldhof fermenter 199
- Water, as media component 96
- Water quality indicators 314
- Weight, measurement of fermenter 221
- Whole broth processing 307
- Yeast
 - biomass 7
 - processes, inoculum development 151
 - production for baking industry 1
 - role in alcoholic fermentation 7
- Yield improvement, secondary metabolites 57